MATHEMATICS

BASIC 3

THE LEARNER’S DISCOVERY GUIDE

39 illustrated lessons | Worked examples | Classroom projects | 585 digital practice questions

Original learner content aligned to the supplied 2019 Ghana standards-based Mathematics curriculum.

Includes companion activities for the offline Maths Quest course.

Welcome, young mathematician!

You will learn to count, compare, add, subtract, understand parts, recognise shapes, measure real objects and explain information in graphs.

How to use this book

Read the colourful lesson ideas. Study the picture and worked examples. Solve the practice questions on paper. Complete the real-life activities safely. Then try the five Play rounds and ten Challenge questions in your matching online course.

This guide follows the uploaded Basic 3 Mathematics curriculum. Where the curriculum calls for hands-on investigation, do the activity with a teacher or adult.

Learning map

NUMBER

Quest 01: Count and estimate to 10,000

Quest 02: Find neighbours on large number charts

Quest 03: Build four-digit numbers

Quest 04: Compare and order numbers to 10,000

Quest 05: Move in opposite directions

Quest 06: Understand positive and negative situations

Quest 07: Read negative numbers to minus ten

Quest 08: Use standard addition and subtraction

Quest 09: Balance equal number statements

Quest 10: Mental strategies within 1,000

Quest 11: Flexible addition strategies

Quest 12: Flexible subtraction strategies

Quest 13: Estimate sums and differences

Quest 14: Explore commutative addition

Quest 15: Multiplication as equal groups

Quest 16: Multiplication arrays

Quest 17: Division by sharing equally

Quest 18: Division by repeated subtraction

Quest 19: Division and multiplication families

Quest 20: Understand unit fractions

Quest 21: Fractions of collections

Quest 22: Compare like-denominator fractions

Quest 23: Use Ghanaian money denominations

ALGEBRA

Quest 24: Investigate growing and shrinking patterns

GEOMETRY & MEASUREMENT

Quest 25: Discover faces, edges and vertices

Quest 26: Draw and recognise angles

Quest 27: Compare angles with a square corner

Quest 28: Identify quadrilaterals

Quest 29: Measure distances on number lines

Quest 30: Metres and centimetres

Quest 31: Choose suitable length units

Quest 32: Estimate and measure perimeter

Quest 33: Grams, kilograms, millilitres and litres

Quest 34: Estimate mass and capacity

Quest 35: Compare the duration of activities

Quest 36: Use calendars to plan events

Quest 37: Convert seconds, minutes and hours

DATA

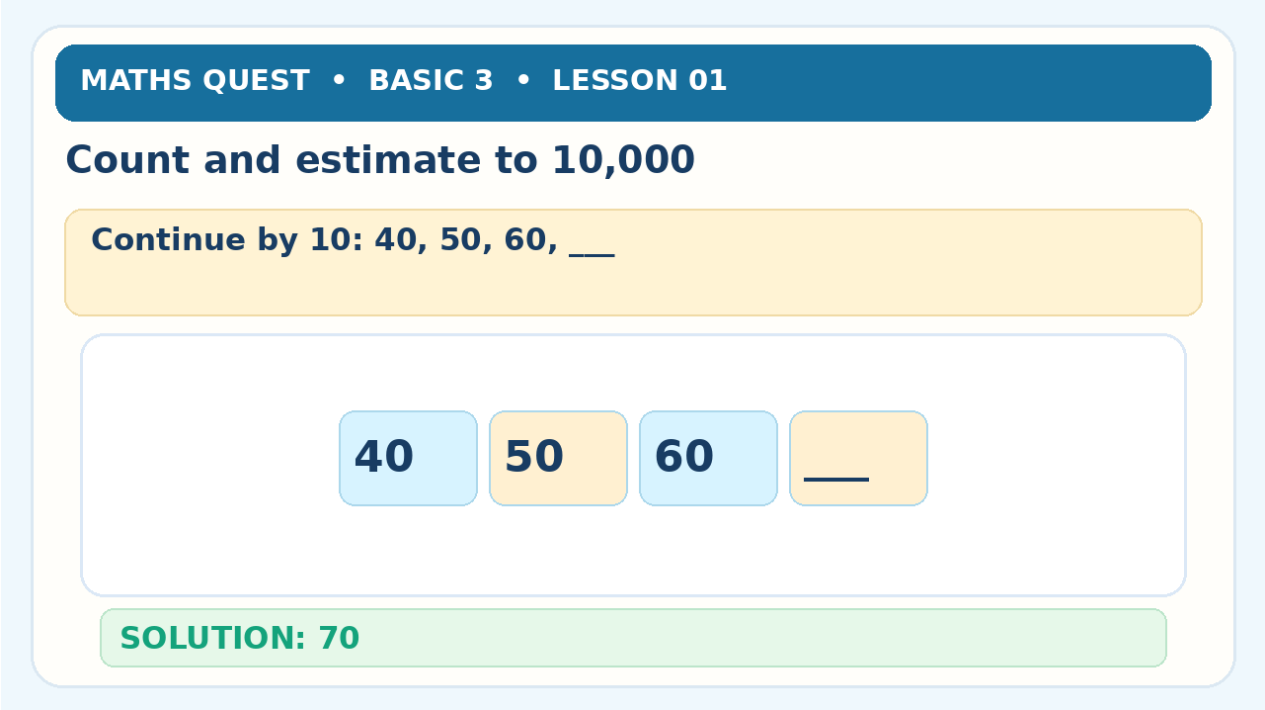
Quest 38: Collect and record useful data

Quest 39: Build and interpret pictographs

Quest 01 | Count and estimate to 10,000

YOUR MISSION

Skip-count forwards and backwards by 10, 50, 100, 500 and 1,000 within 10,000.



LEARN IT - FOUR IMPORTANT IDEAS

1. Large numbers are easier to count in equal jumps. A jump of 100 changes the hundreds digit, while a jump of 1,000 changes the thousands digit.

2. Estimate using familiar groups such as bundles of ten or hundreds, then count to check. The rule of a sequence must stay the same.

3. Worked example: Continue by 10: 40, 50, 60, ___ Answer: 70. Add 10 three times from 40.

4. Check your understanding with another example: Continue by 50: 100, 150, 200, ___ The answer is 250. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Continue by 10: 40, 50, 60, ___	70 — Add 10 three times from 40.
Continue by 50: 100, 150, 200, ___	250 — Add 50 three times from 100.
Continue by 100: 200, 300, 400, ___	500 — Add 100 three times from 200.

TRY IT YOURSELF

P1. Continue by 10: 40, 50, 60, ___ Answer: _____

P2. Continue by 50: 100, 150, 200, ___ Answer: _____

P3. Continue by 100: 200, 300, 400, ___ Answer: _____

P4. Continue by 500: 0, 500, 1000, ___ Answer: _____

P5. Continue by 1000: 0, 1000, 2000, ___ Answer: _____

CHALLENGE PRACTICE

C1. Continue by 10: 1040, 1050, 1060, ___ Answer: _____

C2. Continue by 50: 1100, 1150, 1200, ___ Answer: _____

C3. Continue by 100: 1200, 1300, 1400, ___ Answer: _____

C4. Continue by 500: 1000, 1500, 2000, ___ Answer: _____

C5. Continue by 1000: 1000, 2000, 3000, ___ Answer: _____

C6. Continue by 10: 2040, 2050, 2060, ___ Answer: _____

C7. Continue by 50: 2100, 2150, 2200, ___ Answer: _____

C8. Continue by 100: 2200, 2300, 2400, ___ Answer: _____

C9. Continue by 500: 2000, 2500, 3000, ___ Answer: _____

C10. Continue by 1000: 2000, 3000, 4000, ___ Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Continue by 100: 200, 300, 400, ___

2. Work with a partner to explain why 1500 answers this: Continue by 500: 0, 500, 1000, ___

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Continue by 1000: 0, 1000, 2000, ___

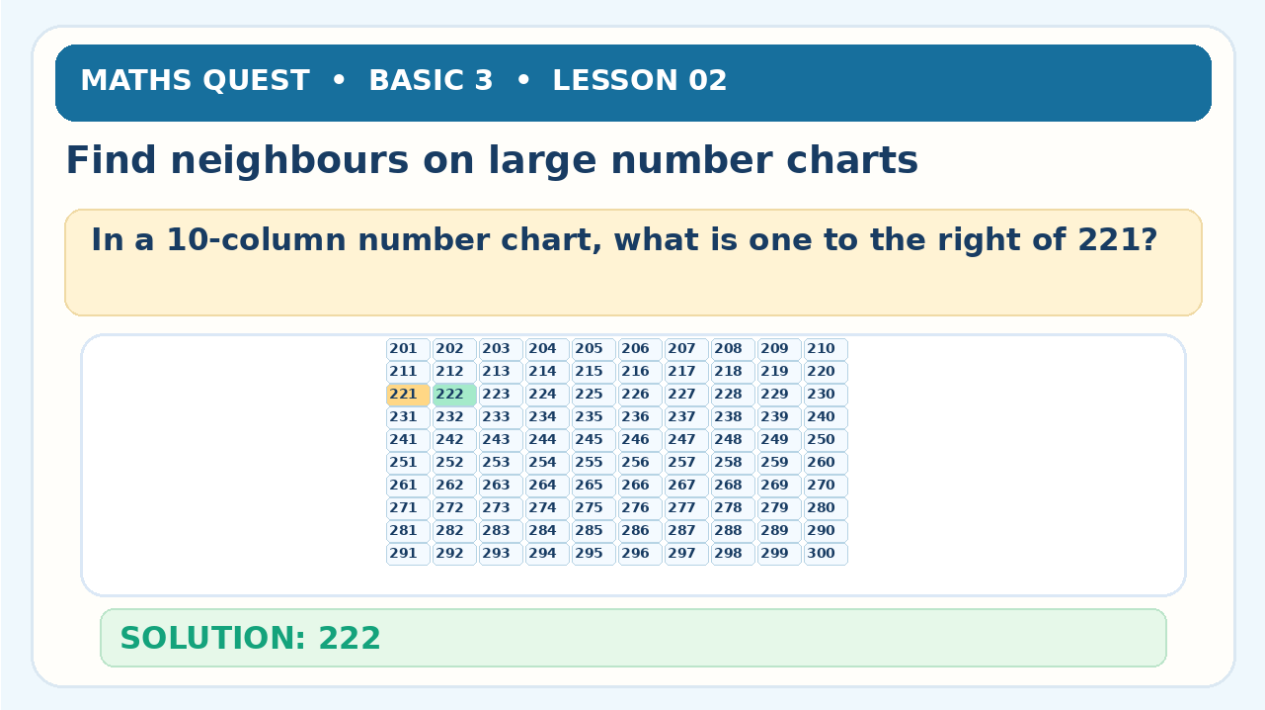
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 02 | Find neighbours on large number charts

YOUR MISSION

Identify numbers above, below, left and right of a target number on a displayed chart.



LEARN IT - FOUR IMPORTANT IDEAS

1. On a chart with ten columns, move right by one and down by ten. Look carefully at the arrangement before deciding.

2. Use your finger to follow exactly one square. A chart may be a small section of a much bigger number chart.

3. Worked example: In a 10-column number chart, what is one to the right of 221? Answer: 222. Each right or left square changes by one; each row changes by ten.

4. Check your understanding with another example: In a 10-column number chart, what is one to the left of 252? The answer is 251. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
In a 10-column number chart, what is one to the right of 221?	222 — Each right or left square changes by one; each row changes by ten.

Try this	How it works
In a 10-column number chart, what is one to the left of 252?	251 — Each right or left square changes by one; each row changes by ten.
In a 10-column number chart, what is one row below of 283?	293 — Each right or left square changes by one; each row changes by ten.

TRY IT YOURSELF

P1. In a 10-column number chart, what is one to the right of 221? Answer: _____

P2. In a 10-column number chart, what is one to the left of 252? Answer: _____

P3. In a 10-column number chart, what is one row below of 283? Answer: _____

P4. In a 10-column number chart, what is one row above of 314? Answer: _____

P5. In a 10-column number chart, what is one to the right of 345? Answer: _____

CHALLENGE PRACTICE

C1. In a 10-column number chart, what is one to the left of 376? Answer: _____

C2. In a 10-column number chart, what is one row below of 407? Answer: _____

C3. In a 10-column number chart, what is one row above of 438? Answer: _____

C4. In a 10-column number chart, what is one to the right of 469? Answer: _____

C5. In a 10-column number chart, what is one to the left of 500? Answer: _____

C6. In a 10-column number chart, what is one row below of 531? Answer: _____

C7. In a 10-column number chart, what is one row above of 562? Answer: _____

C8. In a 10-column number chart, what is one to the right of 593? Answer: _____

C9. In a 10-column number chart, what is one to the left of 624? Answer: _____

C10. In a 10-column number chart, what is one row below of 655? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: In a 10-column number chart, what is one row below of 283?
2. Work with a partner to explain why 304 answers this: In a 10-column number chart, what is one row above of 314?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: In a 10-column number chart, what is one to the right of 345?

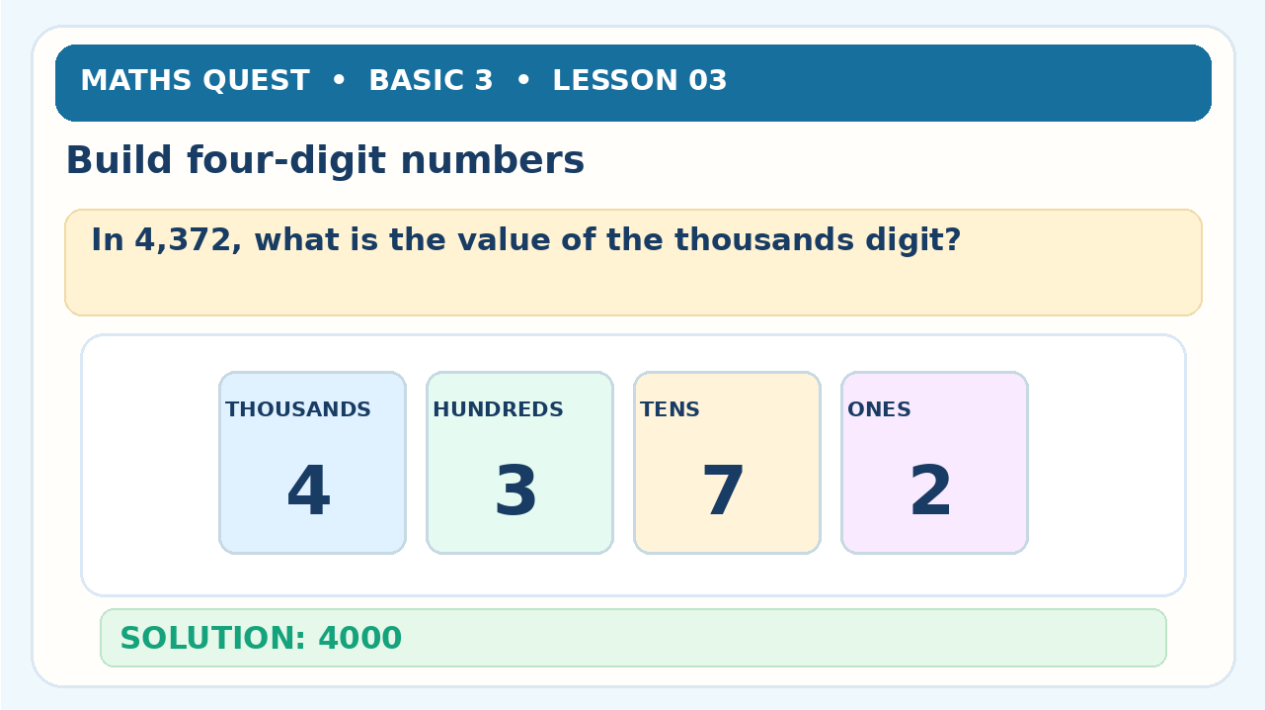
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 03 | Build four-digit numbers

YOUR MISSION

Represent thousands, hundreds, tens and ones in several equivalent ways.



LEARN IT - FOUR IMPORTANT IDEAS

1. In 4,372, the 4 means four thousands and the 3 means three hundreds. The value of a digit comes from its position.

2. Expanded notation shows each place: 4,372 = 4,000 + 300 + 70 + 2. Draw place-value blocks or a four-column mat.

3. Worked example: In 4,372, what is the value of the thousands digit? Answer: 4000. The digit in that place is worth 4,000.

4. Check your understanding with another example: In 6,805, what is the value of the hundreds digit? The answer is 800. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
In 4,372, what is the value of the thousands digit?	4000 — The digit in that place is worth 4,000.
In 6,805, what is the value of the hundreds digit?	800 — The digit in that place is worth 800.

Try this	How it works
In 9,134, what is the value of the tens digit?	30 — The digit in that place is worth 30.

TRY IT YOURSELF

P1. In 4,372, what is the value of the thousands digit? Answer: _____

P2. In 6,805, what is the value of the hundreds digit? Answer: _____

P3. In 9,134, what is the value of the tens digit? Answer: _____

P4. In 2,057, what is the value of the ones digit? Answer: _____

P5. In 7,481, what is the value of the thousands digit? Answer: _____

CHALLENGE PRACTICE

C1. In 3,260, what is the value of the hundreds digit? Answer: _____

C2. In 5,402, what is the value of the tens digit? Answer: _____

C3. In 8,649, what is the value of the ones digit? Answer: _____

C4. In 1,307, what is the value of the thousands digit? Answer: _____

C5. In 9,991, what is the value of the hundreds digit? Answer: _____

C6. In 4,705, what is the value of the tens digit? Answer: _____

C7. In 7,213, what is the value of the ones digit? Answer: _____

C8. In 6,060, what is the value of the thousands digit? Answer: _____

C9. In 2,845, what is the value of the hundreds digit? Answer: _____

C10. In 1,589, what is the value of the tens digit? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: In 9,134, what is the value of the tens digit?

2. Work with a partner to explain why 7 answers this: In 2,057, what is the value of the ones digit?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: In 7,481, what is the value of the thousands digit?

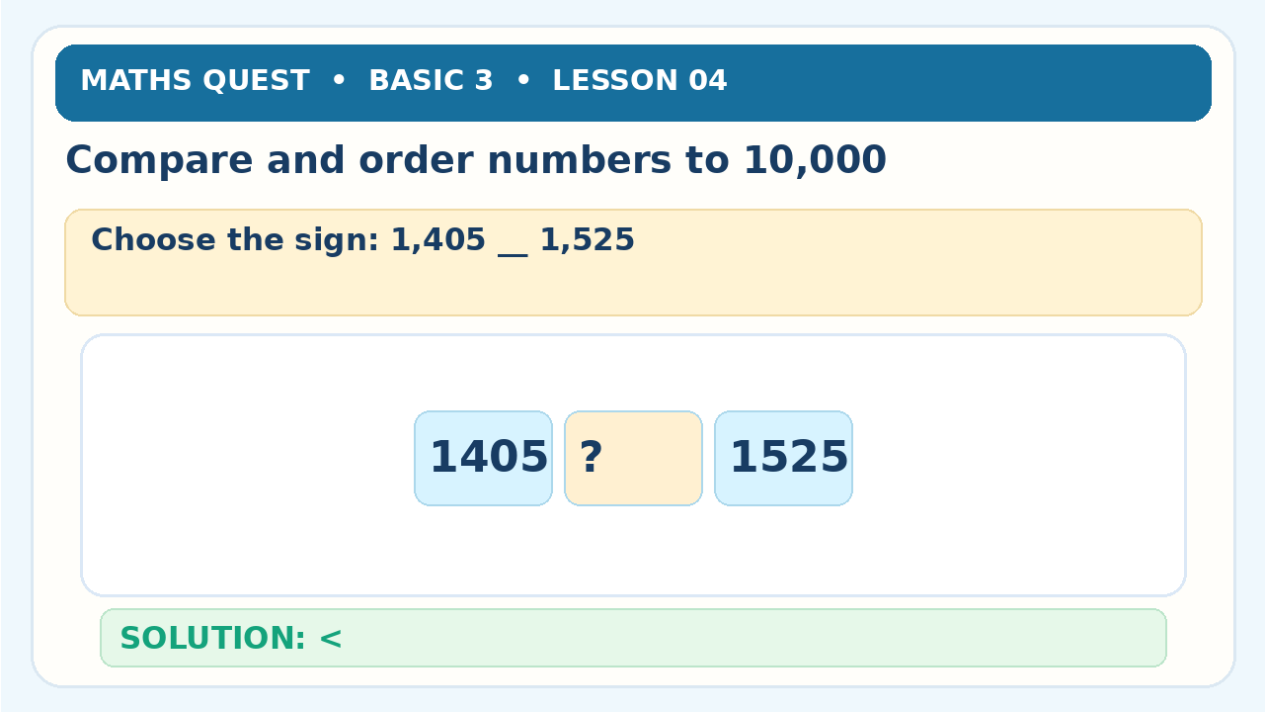
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 04 | Compare and order numbers to 10,000

YOUR MISSION

Compare and order whole numbers using thousands, hundreds, tens and ones.



LEARN IT - FOUR IMPORTANT IDEAS

1. Start comparing from the leftmost digit. If the thousands match, compare hundreds, then tens and ones.

2. Use the signs < for smaller, > for larger and = for equal. To order several numbers, compare systematically.

3. Worked example: Choose the sign: 1,405 __ 1,525 Answer: <. Compare the thousands, hundreds, tens and ones.

4. Check your understanding with another example: Choose the sign: 1,596 __ 1,476 The answer is >. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Choose the sign: 1,405 __ 1,525	< — Compare the thousands, hundreds, tens and ones.

Try this	How it works
Choose the sign: 1,596 __ 1,476	> — Compare the thousands, hundreds, tens and ones.
Choose the sign: 1,787 __ 1,787	= — Compare the thousands, hundreds, tens and ones.

TRY IT YOURSELF

P1. Choose the sign: 1,405 __ 1,525 Answer: _____

P2. Choose the sign: 1,596 __ 1,476 Answer: _____

P3. Choose the sign: 1,787 __ 1,787 Answer: _____

P4. Choose the sign: 1,978 __ 1,990 Answer: _____

P5. Choose the sign: 2,169 __ 2,157 Answer: _____

CHALLENGE PRACTICE

C1. Choose the sign: 2,360 __ 2,480 Answer: _____

C2. Choose the sign: 2,551 __ 2,431 Answer: _____

C3. Choose the sign: 2,742 __ 2,742 Answer: _____

C4. Choose the sign: 2,933 __ 2,945 Answer: _____

C5. Choose the sign: 3,124 __ 3,112 Answer: _____

C6. Choose the sign: 3,315 __ 3,435 Answer: _____

C7. Choose the sign: 3,506 __ 3,386 Answer: _____

C8. Choose the sign: 3,697 __ 3,697 Answer: _____

C9. Choose the sign: 3,888 __ 3,900 Answer: _____

C10. Choose the sign: 4,079 __ 4,067 Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Choose the sign: 1,787 __ 1,787

2. Work with a partner to explain why < answers this: Choose the sign: 1,978 __ 1,990

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Choose the sign: 2,169 __ 2,157

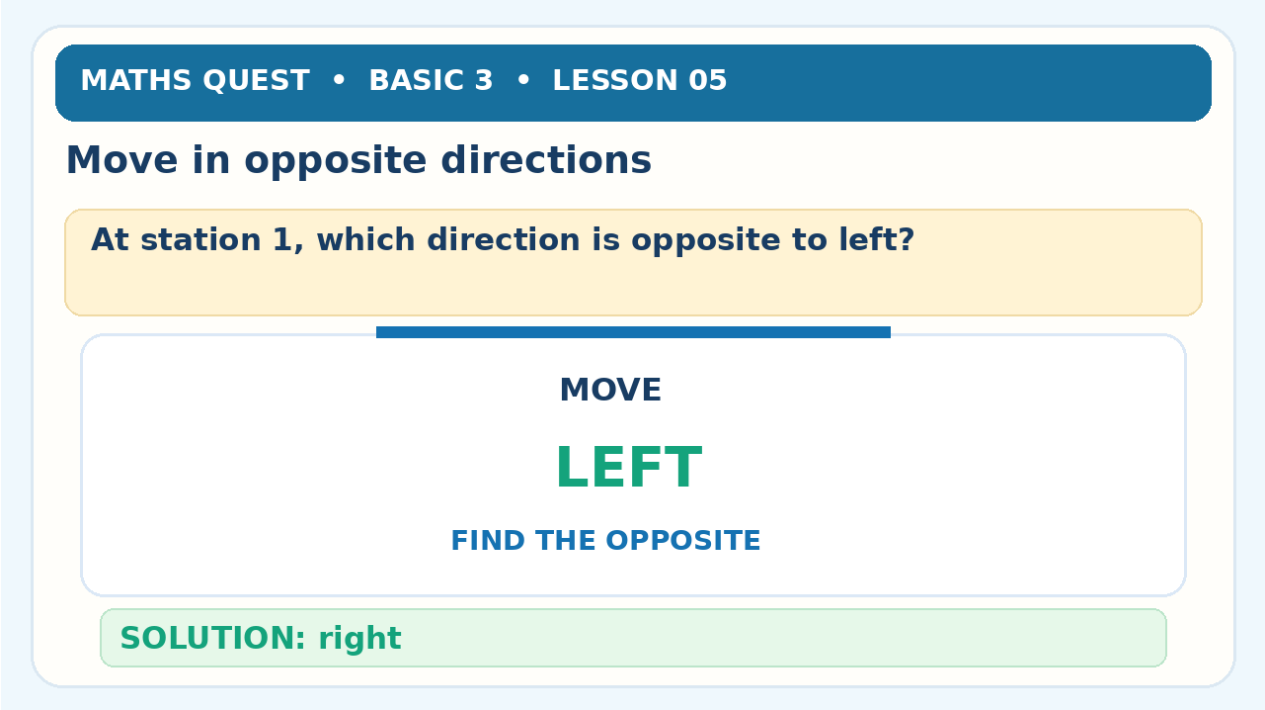
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 05 | Move in opposite directions

YOUR MISSION

Recognise and describe movements and values that are opposites.



LEARN IT - FOUR IMPORTANT IDEAS

1. Moving right and moving left are opposite directions. Going up and going down are also opposites.

2. Choose a starting position and describe each move. Two pupils can move the same distance in opposite directions.

3. Worked example: At station 1, which direction is opposite to left? Answer: right. Opposites point in reverse directions.

4. Check your understanding with another example: At station 2, which direction is opposite to up? The answer is down. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
At station 1, which direction is opposite to left?	right — Opposites point in reverse directions.
At station 2, which direction is opposite to up?	down — Opposites point in reverse directions.

Try this	How it works
At station 3, which direction is opposite to forward?	backward — Opposites point in reverse directions.

TRY IT YOURSELF

P1. At station 1, which direction is opposite to left? Answer: _____

P2. At station 2, which direction is opposite to up? Answer: _____

P3. At station 3, which direction is opposite to forward? Answer: _____

P4. At station 4, which direction is opposite to north? Answer: _____

P5. At station 5, which direction is opposite to east? Answer: _____

CHALLENGE PRACTICE

C1. At station 6, which direction is opposite to right? Answer: _____

C2. At station 7, which direction is opposite to down? Answer: _____

C3. At station 8, which direction is opposite to backward? Answer: _____

C4. At station 9, which direction is opposite to south? Answer: _____

C5. At station 10, which direction is opposite to west? Answer: _____

C6. At station 11, which direction is opposite to left? Answer: _____

C7. At station 12, which direction is opposite to up? Answer: _____

C8. At station 13, which direction is opposite to forward? Answer: _____

C9. At station 14, which direction is opposite to north? Answer: _____

C10. At station 15, which direction is opposite to east? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: At station 3, which direction is opposite to forward?
2. Work with a partner to explain why south answers this: At station 4, which direction is opposite to north?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: At station 5, which direction is opposite to east?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 06 | Understand positive and negative situations

YOUR MISSION

Use sea level and other real settings to make sense of positive and negative values.

MATHS QUEST • BASIC 3 • LESSON 06

Understand positive and negative situations

Sea level marker 1: a point is 1 metres above sea level. Write its signed height.



SOLUTION: 1

LEARN IT - FOUR IMPORTANT IDEAS

1. Sea level can be marked zero. Above sea level can be represented by positive values, and below by negative values.

2. A temperature below zero can be written with a minus sign. The sign tells the side of zero, not just the size.

3. Worked example: Sea level marker 1: a point is 1 metres above sea level. Write its signed height. Answer: 1. Above zero is positive; below zero is negative.

4. Check your understanding with another example: Sea level marker 2: a point is 2 metres below sea level. Write its signed height. The answer is -2. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Sea level marker 1: a point is 1 metres above sea level. Write its signed height.	1 — Above zero is positive; below zero is negative.

Try this	How it works
Sea level marker 2: a point is 2 metres below sea level. Write its signed height.	-2 — Above zero is positive; below zero is negative.
Sea level marker 3: a point is 3 metres above sea level. Write its signed height.	3 — Above zero is positive; below zero is negative.

TRY IT YOURSELF

P1. Sea level marker 1: a point is 1 metres above sea level. Write its signed height. Answer: _____

P2. Sea level marker 2: a point is 2 metres below sea level. Write its signed height. Answer: _____

P3. Sea level marker 3: a point is 3 metres above sea level. Write its signed height. Answer: _____

P4. Sea level marker 4: a point is 4 metres below sea level. Write its signed height. Answer: _____

P5. Sea level marker 5: a point is 5 metres above sea level. Write its signed height. Answer: _____

CHALLENGE PRACTICE

C1. Sea level marker 6: a point is 6 metres below sea level. Write its signed height. Answer: _____

C2. Sea level marker 7: a point is 7 metres above sea level. Write its signed height. Answer: _____

C3. Sea level marker 8: a point is 8 metres below sea level. Write its signed height. Answer: _____

C4. Sea level marker 9: a point is 9 metres above sea level. Write its signed height. Answer: _____

C5. Sea level marker 10: a point is 10 metres below sea level. Write its signed height. Answer: _____

C6. Sea level marker 11: a point is 1 metres above sea level. Write its signed height. Answer: _____

C7. Sea level marker 12: a point is 2 metres below sea level. Write its signed height. Answer: _____

C8. Sea level marker 13: a point is 3 metres above sea level. Write its signed height. Answer: _____

C9. Sea level marker 14: a point is 4 metres below sea level. Write its signed height. Answer: _____

C10. Sea level marker 15: a point is 5 metres above sea level. Write its signed height. Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Sea level marker 3: a point is 3 metres above sea level. Write its signed height.

2. Work with a partner to explain why -4 answers this: Sea level marker 4: a point is 4 metres below sea level. Write its signed height.

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Sea level marker 5: a point is 5 metres above sea level. Write its signed height.

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 07 | Read negative numbers to minus ten

YOUR MISSION

Locate negative values between minus ten and zero on a number line.

MATHS QUEST • BASIC 3 • LESSON 07

Read negative numbers to minus ten

At point 1, which integer comes immediately to the right of -10 on a number line?



SOLUTION: -9

LEARN IT - FOUR IMPORTANT IDEAS

1. Negative numbers are placed to the left of zero on a horizontal number line. Moving left makes the value smaller.

2. Minus three is to the right of minus five, so minus three is the greater number. Practise hopping from zero.

3. Worked example: At point 1, which integer comes immediately to the right of -10 on a number line? Answer: -9. Move right by one, even for negative numbers.

4. Check your understanding with another example: At point 2, which integer comes immediately to the right of -9 on a number line? The answer is -8. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
At point 1, which integer comes immediately to the right of -10 on a number line?	-9 — Move right by one, even for negative numbers.

Try this	How it works
At point 2, which integer comes immediately to the right of -9 on a number line?	-8 — Move right by one, even for negative numbers.
At point 3, which integer comes immediately to the right of -8 on a number line?	-7 — Move right by one, even for negative numbers.

TRY IT YOURSELF

P1. At point 1, which integer comes immediately to the right of -10 on a number line? Answer: _____

P2. At point 2, which integer comes immediately to the right of -9 on a number line? Answer: _____

P3. At point 3, which integer comes immediately to the right of -8 on a number line? Answer: _____

P4. At point 4, which integer comes immediately to the right of -7 on a number line? Answer: _____

P5. At point 5, which integer comes immediately to the right of -6 on a number line? Answer: _____

CHALLENGE PRACTICE

C1. At point 6, which integer comes immediately to the right of -5 on a number line? Answer: _____

C2. At point 7, which integer comes immediately to the right of -4 on a number line? Answer: _____

C3. At point 8, which integer comes immediately to the right of -3 on a number line? Answer: _____

C4. At point 9, which integer comes immediately to the right of -2 on a number line? Answer: _____

C5. At point 10, which integer comes immediately to the right of -1 on a number line? Answer: _____

C6. At point 11, which integer comes immediately to the right of -10 on a number line? Answer: _____

C7. At point 12, which integer comes immediately to the right of -9 on a number line? Answer: _____

C8. At point 13, which integer comes immediately to the right of -8 on a number line? Answer: _____

C9. At point 14, which integer comes immediately to the right of -7 on a number line? Answer: _____

C10. At point 15, which integer comes immediately to the right of -6 on a number line? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: At point 3, which integer comes immediately to the right of -8 on a number line?

2. Work with a partner to explain why -6 answers this: At point 4, which integer comes immediately to the right of -7 on a number line?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: At point 5, which integer comes immediately to the right of -6 on a number line?

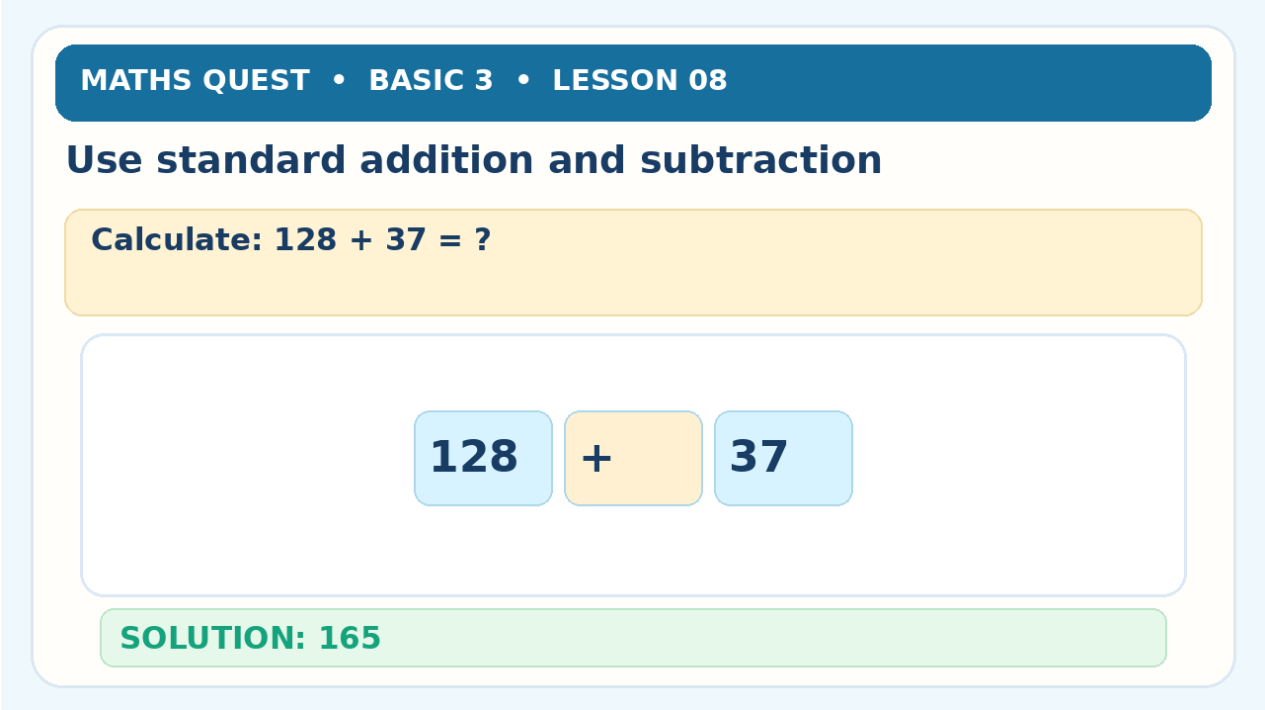
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 08 | Use standard addition and subtraction

YOUR MISSION

Solve additions and subtractions within 1,000, including unknown numbers.



LEARN IT - FOUR IMPORTANT IDEAS

1. Line up hundreds, tens and ones. Regroup when you have ten ones or need to exchange one ten.

2. Always check subtraction with addition. If 452 minus 186 equals 266, then 266 plus 186 equals 452.

3. Worked example: Calculate: 128 + 37 = ? Answer: 165. Add hundreds, tens and ones; regroup when needed.

4. Check your understanding with another example: Calculate: 157 - 50 = ? The answer is 107. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Calculate: 128 + 37 = ?	165 — Add hundreds, tens and ones; regroup when needed.

Try this	How it works
Calculate: 157 - 50 = ?	107 — 157 minus 50 is 107. Check by adding 50.
Calculate: 186 + 63 = ?	249 — Add hundreds, tens and ones; regroup when needed.

TRY IT YOURSELF

P1. Calculate: 128 + 37 = ? Answer: _____

P2. Calculate: 157 - 50 = ? Answer: _____

P3. Calculate: 186 + 63 = ? Answer: _____

P4. Calculate: 215 - 76 = ? Answer: _____

P5. Calculate: 244 + 89 = ? Answer: _____

CHALLENGE PRACTICE

C1. Calculate: 273 - 102 = ? Answer: _____

C2. Calculate: 302 + 115 = ? Answer: _____

C3. Calculate: 331 - 37 = ? Answer: _____

C4. Calculate: 360 + 50 = ? Answer: _____

C5. Calculate: 389 - 63 = ? Answer: _____

C6. Calculate: 418 + 76 = ? Answer: _____

C7. Calculate: 447 - 89 = ? Answer: _____

C8. Calculate: 476 + 102 = ? Answer: _____

C9. Calculate: 505 - 115 = ? Answer: _____

C10. Calculate: 534 + 37 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Calculate: 186 + 63 = ?

2. Work with a partner to explain why 139 answers this: Calculate: 215 - 76 = ?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Calculate: 244 + 89 = ?

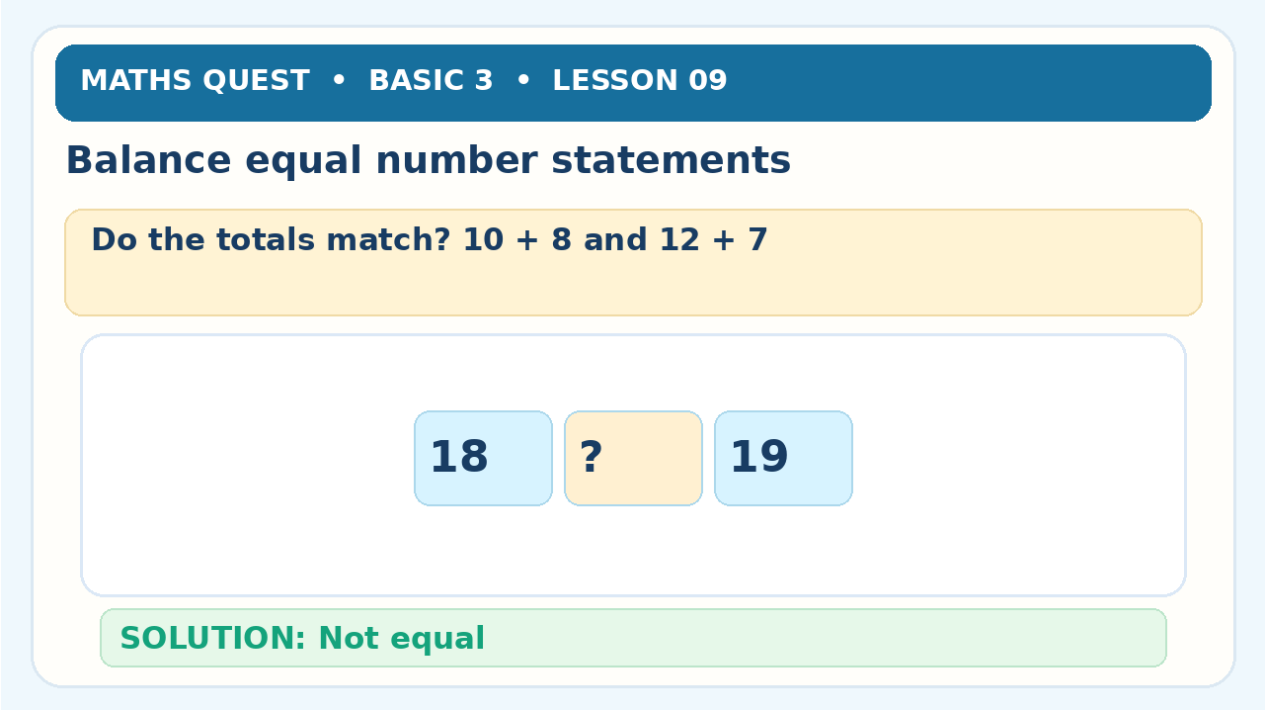
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 09 | Balance equal number statements

YOUR MISSION

Identify equal and unequal expressions and solve missing-number equations.



LEARN IT - FOUR IMPORTANT IDEAS

1. An equals sign means both sides have the same total. Calculate each side before comparing.

2. If a box hides a value, use the inverse operation to restore balance. For example, 7 + box = 12 gives box = 5.

3. Worked example: Do the totals match? 10 + 8 and 12 + 7 Answer: Not equal. Check both totals before choosing a sign.

4. Check your understanding with another example: Do the totals match? 11 + 9 and 13 + 7 The answer is Equal. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Do the totals match? 10 + 8 and 12 + 7	Not equal — Check both totals before choosing a sign.
Do the totals match? 11 + 9 and 13 + 7	Equal — Check both totals before choosing a sign.

Try this	How it works
Do the totals match? 12 + 10 and 14 + 8	Equal — Check both totals before choosing a sign.

TRY IT YOURSELF

P1. Do the totals match? 10 + 8 and 12 + 7 Answer: _____

P2. Do the totals match? 11 + 9 and 13 + 7 Answer: _____

P3. Do the totals match? 12 + 10 and 14 + 8 Answer: _____

P4. Do the totals match? 13 + 11 and 15 + 10 Answer: _____

P5. Do the totals match? 14 + 12 and 16 + 10 Answer: _____

CHALLENGE PRACTICE

C1. Do the totals match? 15 + 13 and 17 + 11 Answer: _____

C2. Do the totals match? 16 + 14 and 18 + 13 Answer: _____

C3. Do the totals match? 17 + 15 and 19 + 13 Answer: _____

C4. Do the totals match? 18 + 16 and 20 + 14 Answer: _____

C5. Do the totals match? 19 + 17 and 21 + 16 Answer: _____

C6. Do the totals match? 20 + 18 and 22 + 16 Answer: _____

C7. Do the totals match? 21 + 19 and 23 + 17 Answer: _____

C8. Do the totals match? 22 + 20 and 24 + 19 Answer: _____

C9. Do the totals match? 23 + 21 and 25 + 19 Answer: _____

C10. Do the totals match? 24 + 22 and 26 + 20 Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Do the totals match? 12 + 10 and 14 + 8
2. Work with a partner to explain why Not equal answers this: Do the totals match? 13 + 11 and 15 + 10

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Do the totals match? 14 + 12 and 16 + 10

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 10 | Mental strategies within 1,000

YOUR MISSION

Add and subtract mentally using hundreds, tens, doubles and friendly numbers.



LEARN IT - FOUR IMPORTANT IDEAS

1. Break numbers into place values to do mental arithmetic. For 240 + 130, add hundreds and tens separately.

2. Use compensation: 199 + 32 can be thought of as 200 + 31. Explain the shortcut to a friend.

3. Worked example: Calculate: 128 + 37 = ? Answer: 165. Add hundreds, tens and ones; regroup when needed.

4. Check your understanding with another example: Calculate: 157 + 50 = ? The answer is 207. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Calculate: 128 + 37 = ?	165 — Add hundreds, tens and ones; regroup when needed.
Calculate: 157 + 50 = ?	207 — Add hundreds, tens and ones; regroup when needed.

Try this	How it works
Calculate: 186 + 63 = ?	249 — Add hundreds, tens and ones; regroup when needed.

TRY IT YOURSELF

P1. Calculate: 128 + 37 = ? Answer: _____

P2. Calculate: 157 + 50 = ? Answer: _____

P3. Calculate: 186 + 63 = ? Answer: _____

P4. Calculate: 215 + 76 = ? Answer: _____

P5. Calculate: 244 + 89 = ? Answer: _____

CHALLENGE PRACTICE

C1. Calculate: 273 + 102 = ? Answer: _____

C2. Calculate: 302 + 115 = ? Answer: _____

C3. Calculate: 331 + 37 = ? Answer: _____

C4. Calculate: 360 + 50 = ? Answer: _____

C5. Calculate: 389 + 63 = ? Answer: _____

C6. Calculate: 418 + 76 = ? Answer: _____

C7. Calculate: 447 + 89 = ? Answer: _____

C8. Calculate: 476 + 102 = ? Answer: _____

C9. Calculate: 505 + 115 = ? Answer: _____

C10. Calculate: 534 + 37 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Calculate: 186 + 63 = ?

2. Work with a partner to explain why 291 answers this: Calculate: 215 + 76 = ?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Calculate: 244 + 89 = ?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 11 | Flexible addition strategies

YOUR MISSION

Use partitioning, friendly jumps and place-value models to add within 1,000.



LEARN IT - FOUR IMPORTANT IDEAS

1. Partition a number into hundreds, tens and ones. Combine like parts before finding the total.

2. Adding 298 + 127 becomes 300 + 125 using compensation. A number-line model can show the same result.

3. Worked example: Calculate: 128 + 37 = ? Answer: 165. Add hundreds, tens and ones; regroup when needed.

4. Check your understanding with another example: Calculate: 157 + 50 = ? The answer is 207. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Calculate: 128 + 37 = ?	165 — Add hundreds, tens and ones; regroup when needed.
Calculate: 157 + 50 = ?	207 — Add hundreds, tens and ones; regroup when needed.

Try this	How it works
Calculate: 186 + 63 = ?	249 — Add hundreds, tens and ones; regroup when needed.

TRY IT YOURSELF

P1. Calculate: 128 + 37 = ? Answer: _____

P2. Calculate: 157 + 50 = ? Answer: _____

P3. Calculate: 186 + 63 = ? Answer: _____

P4. Calculate: 215 + 76 = ? Answer: _____

P5. Calculate: 244 + 89 = ? Answer: _____

CHALLENGE PRACTICE

C1. Calculate: 273 + 102 = ? Answer: _____

C2. Calculate: 302 + 115 = ? Answer: _____

C3. Calculate: 331 + 37 = ? Answer: _____

C4. Calculate: 360 + 50 = ? Answer: _____

C5. Calculate: 389 + 63 = ? Answer: _____

C6. Calculate: 418 + 76 = ? Answer: _____

C7. Calculate: 447 + 89 = ? Answer: _____

C8. Calculate: 476 + 102 = ? Answer: _____

C9. Calculate: 505 + 115 = ? Answer: _____

C10. Calculate: 534 + 37 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Calculate: 186 + 63 = ?

2. Work with a partner to explain why 291 answers this: Calculate: 215 + 76 = ?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Calculate: 244 + 89 = ?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 12 | Flexible subtraction strategies

YOUR MISSION

Subtract within 1,000 with number lines, regrouping and personal strategies.



LEARN IT - FOUR IMPORTANT IDEAS

1. Break the number being subtracted into smaller parts, or regroup in columns when needed.

2. For 502 - 178, count up from 178 to 502 as a check. Both methods must give the same difference.

3. Worked example: Calculate: 128 - 37 = ? Answer: 91. 128 minus 37 is 91. Check by adding 37.

4. Check your understanding with another example: Calculate: 157 - 50 = ? The answer is 107. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Calculate: 128 - 37 = ?	91 — 128 minus 37 is 91. Check by adding 37.
Calculate: 157 - 50 = ?	107 — 157 minus 50 is 107. Check by adding 50.
Calculate: 186 - 63 = ?	123 — 186 minus 63 is 123. Check by adding 63.

TRY IT YOURSELF

P1. Calculate: 128 - 37 = ? Answer: _____

P2. Calculate: 157 - 50 = ? Answer: _____

P3. Calculate: 186 - 63 = ? Answer: _____

P4. Calculate: 215 - 76 = ? Answer: _____

P5. Calculate: 244 - 89 = ? Answer: _____

CHALLENGE PRACTICE

C1. Calculate: 273 - 102 = ? Answer: _____

C2. Calculate: 302 - 115 = ? Answer: _____

C3. Calculate: 331 - 37 = ? Answer: _____

C4. Calculate: 360 - 50 = ? Answer: _____

C5. Calculate: 389 - 63 = ? Answer: _____

C6. Calculate: 418 - 76 = ? Answer: _____

C7. Calculate: 447 - 89 = ? Answer: _____

C8. Calculate: 476 - 102 = ? Answer: _____

C9. Calculate: 505 - 115 = ? Answer: _____

C10. Calculate: 534 - 37 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Calculate: 186 - 63 = ?

2. Work with a partner to explain why 139 answers this: Calculate: 215 - 76 = ?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Calculate: 244 - 89 = ?

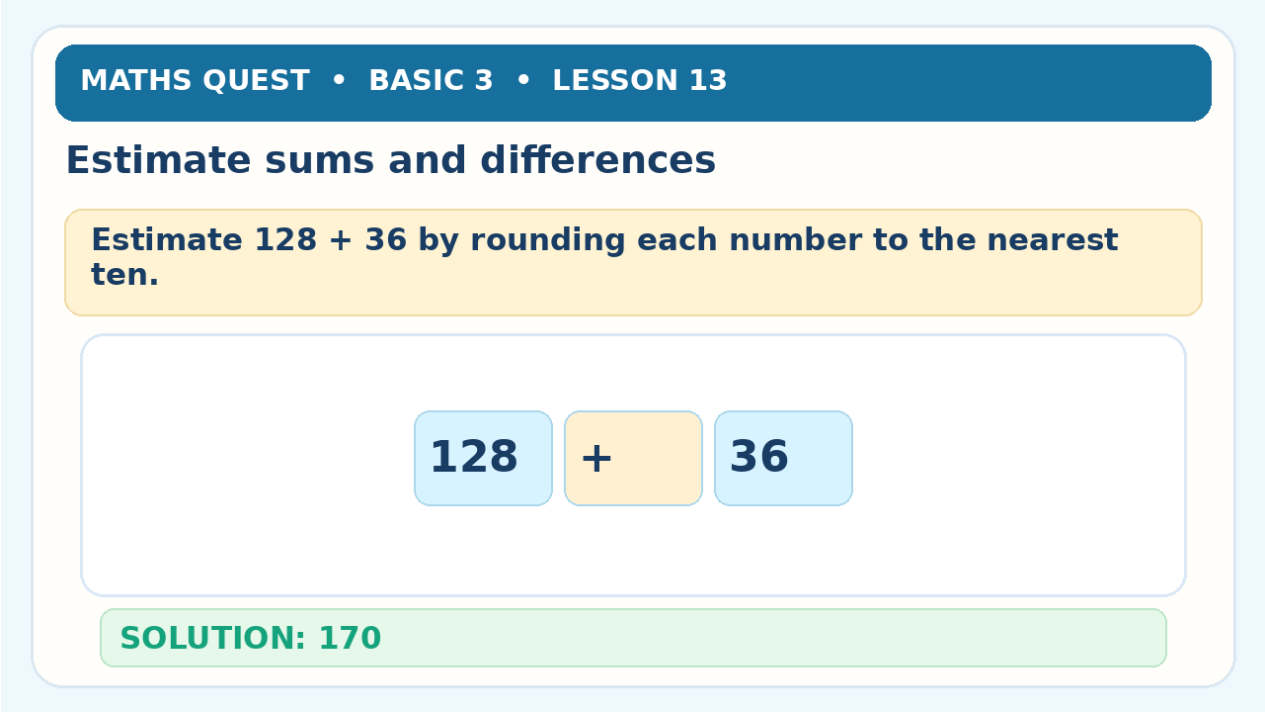
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 13 | Estimate sums and differences

YOUR MISSION

Use rounding to estimate whether an addition or subtraction answer is sensible.



LEARN IT - FOUR IMPORTANT IDEAS

1. Round numbers to nearby hundreds or tens before calculating. An estimate is approximate, not the exact answer.

2. For 348 + 257, round to 350 + 260 to estimate about 610. Compare the actual calculation with the estimate.

3. Worked example: Estimate 128 + 36 by rounding each number to the nearest ten. Answer: 170. 128 rounds to 130 and 36 to 40.

4. Check your understanding with another example: Estimate 158 + 56 by rounding each number to the nearest ten. The answer is 220. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Estimate 128 + 36 by rounding each number to the nearest ten.	170 — 128 rounds to 130 and 36 to 40.
Estimate 158 + 56 by rounding each number to the nearest ten.	220 — 158 rounds to 160 and 56 to 60.

Try this	How it works
Estimate 188 + 66 by rounding each number to the nearest ten.	260 — 188 rounds to 190 and 66 to 70.

TRY IT YOURSELF

P1. Estimate 128 + 36 by rounding each number to the nearest ten. Answer: _____

P2. Estimate 158 + 56 by rounding each number to the nearest ten. Answer: _____

P3. Estimate 188 + 66 by rounding each number to the nearest ten. Answer: _____

P4. Estimate 218 + 76 by rounding each number to the nearest ten. Answer: _____

P5. Estimate 248 + 86 by rounding each number to the nearest ten. Answer: _____

CHALLENGE PRACTICE

C1. Estimate 278 + 106 by rounding each number to the nearest ten. Answer: _____

C2. Estimate 308 + 116 by rounding each number to the nearest ten. Answer: _____

C3. Estimate 338 + 36 by rounding each number to the nearest ten. Answer: _____

C4. Estimate 368 + 56 by rounding each number to the nearest ten. Answer: _____

C5. Estimate 388 + 66 by rounding each number to the nearest ten. Answer: _____

C6. Estimate 418 + 76 by rounding each number to the nearest ten. Answer: _____

C7. Estimate 448 + 86 by rounding each number to the nearest ten. Answer: _____

C8. Estimate 478 + 106 by rounding each number to the nearest ten. Answer: _____

C9. Estimate 508 + 116 by rounding each number to the nearest ten. Answer: _____

C10. Estimate 538 + 36 by rounding each number to the nearest ten. Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Estimate 188 + 66 by rounding each number to the nearest ten.
2. Work with a partner to explain why 300 answers this: Estimate 218 + 76 by rounding each number to the nearest ten.

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Estimate 248 + 86 by rounding each number to the nearest ten.

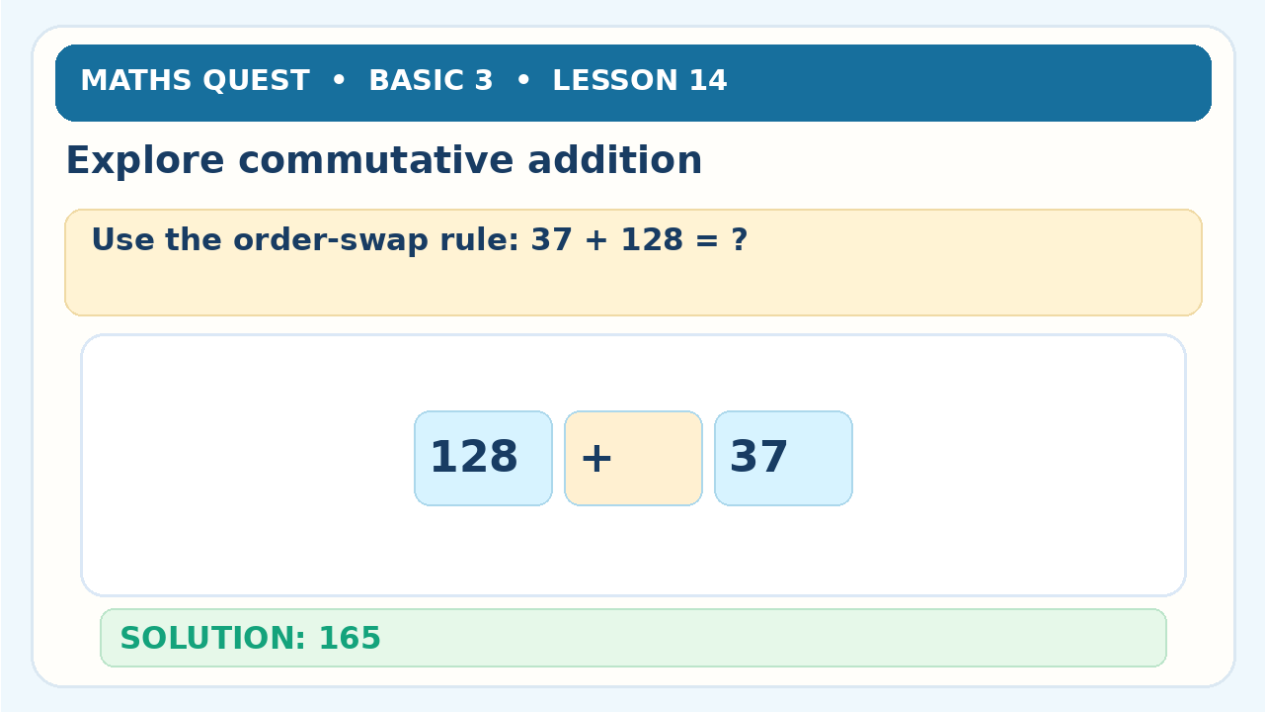
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 14 | Explore commutative addition

YOUR MISSION

Demonstrate that changing the order of addends does not change the sum.



LEARN IT - FOUR IMPORTANT IDEAS

1. When two groups join, you can switch their order and still have the same total. This is the commutative property of addition.

2. For example 27 + 14 = 14 + 27. Show the idea with counters or two differently ordered number sentences.

3. Worked example: Use the order-swap rule: 37 + 128 = ? Answer: 165. The order of two addends does not change the sum.

4. Check your understanding with another example: Use the order-swap rule: 50 + 157 = ? The answer is 207. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Use the order-swap rule: 37 + 128 = ?	165 — The order of two addends does not change the sum.
Use the order-swap rule: 50 + 157 = ?	207 — The order of two addends does not change the sum.

Try this	How it works
Use the order-swap rule: 63 + 186 = ?	249 — The order of two addends does not change the sum.

TRY IT YOURSELF

P1. Use the order-swap rule: 37 + 128 = ? Answer: _____

P2. Use the order-swap rule: 50 + 157 = ? Answer: _____

P3. Use the order-swap rule: 63 + 186 = ? Answer: _____

P4. Use the order-swap rule: 76 + 215 = ? Answer: _____

P5. Use the order-swap rule: 89 + 244 = ? Answer: _____

CHALLENGE PRACTICE

C1. Use the order-swap rule: 102 + 273 = ? Answer: _____

C2. Use the order-swap rule: 115 + 302 = ? Answer: _____

C3. Use the order-swap rule: 37 + 331 = ? Answer: _____

C4. Use the order-swap rule: 50 + 360 = ? Answer: _____

C5. Use the order-swap rule: 63 + 389 = ? Answer: _____

C6. Use the order-swap rule: 76 + 418 = ? Answer: _____

C7. Use the order-swap rule: 89 + 447 = ? Answer: _____

C8. Use the order-swap rule: 102 + 476 = ? Answer: _____

C9. Use the order-swap rule: 115 + 505 = ? Answer: _____

C10. Use the order-swap rule: 37 + 534 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Use the order-swap rule: 63 + 186 = ?

2. Work with a partner to explain why 291 answers this: Use the order-swap rule: 76 + 215 = ?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Use the order-swap rule: 89 + 244 = ?

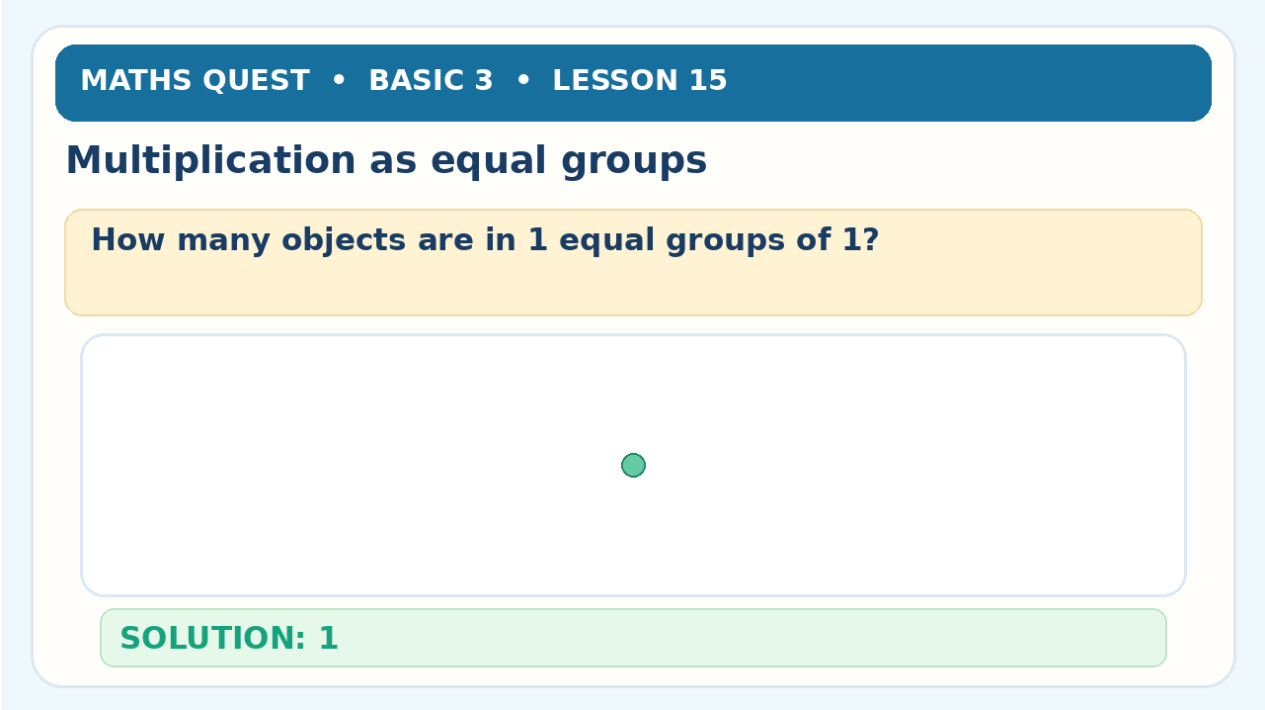
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 15 | Multiplication as equal groups

YOUR MISSION

Use groups and skip counting to multiply numbers to 5 by 5.



LEARN IT - FOUR IMPORTANT IDEAS

1. Multiplication is a short way to count equal groups. Three groups of four make twelve.

2. Write 3 × 4 = 12 and draw three groups, each with four counters. Check with repeated addition.

3. Worked example: How many objects are in 1 equal groups of 1? Answer: 1. Count 1 groups of 1: 1 × 1 = 1.

4. Check your understanding with another example: How many objects are in 2 equal groups of 2? The answer is 4. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
How many objects are in 1 equal groups of 1?	1 — Count 1 groups of 1: 1 × 1 = 1.
How many objects are in 2 equal groups of 2?	4 — Count 2 groups of 2: 2 × 2 = 4.
How many objects are in 3 equal groups of 3?	9 — Count 3 groups of 3: 3 × 3 = 9.

TRY IT YOURSELF

P1. How many objects are in 1 equal groups of 1? Answer: _____

P2. How many objects are in 2 equal groups of 2? Answer: _____

P3. How many objects are in 3 equal groups of 3? Answer: _____

P4. How many objects are in 4 equal groups of 4? Answer: _____

P5. How many objects are in 5 equal groups of 5? Answer: _____

CHALLENGE PRACTICE

C1. How many objects are in 1 equal groups of 2? Answer: _____

C2. How many objects are in 2 equal groups of 3? Answer: _____

C3. How many objects are in 3 equal groups of 4? Answer: _____

C4. How many objects are in 4 equal groups of 5? Answer: _____

C5. How many objects are in 5 equal groups of 1? Answer: _____

C6. How many objects are in 1 equal groups of 3? Answer: _____

C7. How many objects are in 2 equal groups of 4? Answer: _____

C8. How many objects are in 3 equal groups of 5? Answer: _____

C9. How many objects are in 4 equal groups of 1? Answer: _____

C10. How many objects are in 5 equal groups of 2? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: How many objects are in 3 equal groups of 3?

2. Work with a partner to explain why 16 answers this: How many objects are in 4 equal groups of 4?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: How many objects are in 5 equal groups of 5?

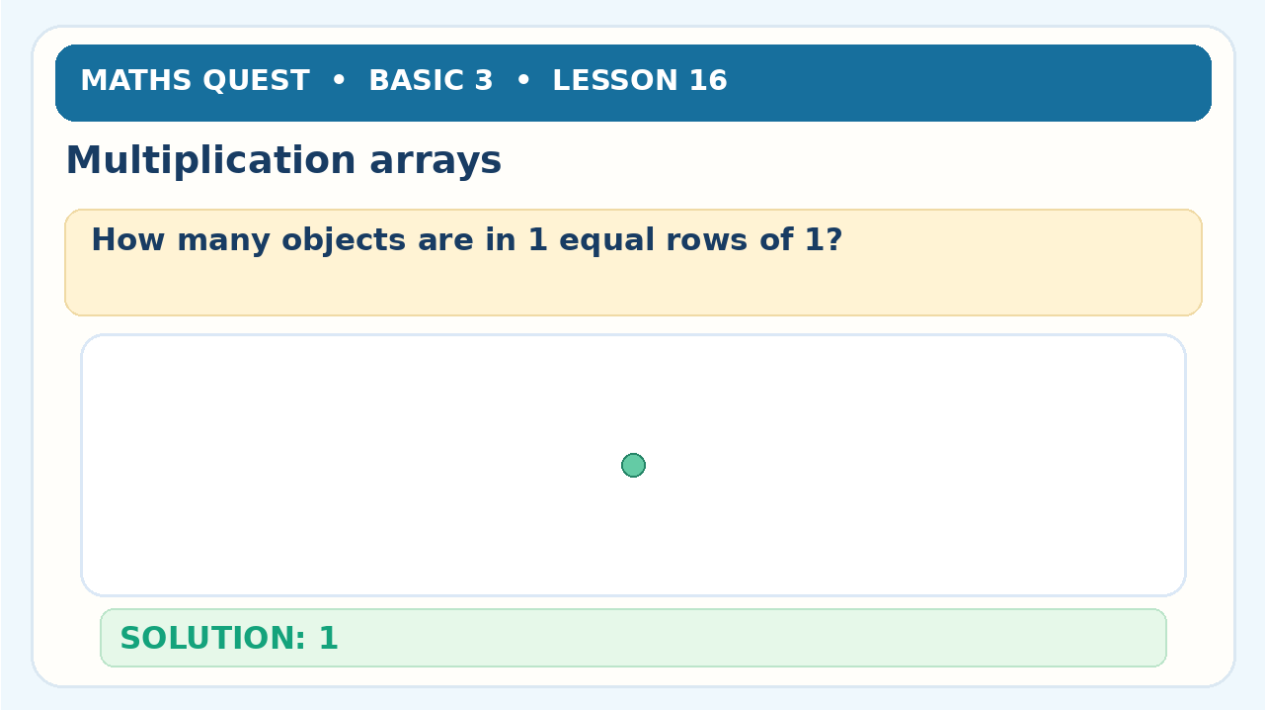
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 16 | Multiplication arrays

YOUR MISSION

Represent multiplication using rows and columns in rectangular arrays.



LEARN IT - FOUR IMPORTANT IDEAS

1. An array puts objects in equal rows and equal columns. Count the rows and objects in each row.

2. Four rows of five dots make twenty. Turn the array and you will still have twenty dots.

3. Worked example: How many objects are in 1 equal rows of 1? Answer: 1. Count 1 groups of 1: 1 × 1 = 1.

4. Check your understanding with another example: How many objects are in 2 equal rows of 2? The answer is 4. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
How many objects are in 1 equal rows of 1?	1 — Count 1 groups of 1: 1 × 1 = 1.
How many objects are in 2 equal rows of 2?	4 — Count 2 groups of 2: 2 × 2 = 4.
How many objects are in 3 equal rows of 3?	9 — Count 3 groups of 3: 3 × 3 = 9.

TRY IT YOURSELF

P1. How many objects are in 1 equal rows of 1? Answer: _____

P2. How many objects are in 2 equal rows of 2? Answer: _____

P3. How many objects are in 3 equal rows of 3? Answer: _____

P4. How many objects are in 4 equal rows of 4? Answer: _____

P5. How many objects are in 5 equal rows of 5? Answer: _____

CHALLENGE PRACTICE

C1. How many objects are in 1 equal rows of 2? Answer: _____

C2. How many objects are in 2 equal rows of 3? Answer: _____

C3. How many objects are in 3 equal rows of 4? Answer: _____

C4. How many objects are in 4 equal rows of 5? Answer: _____

C5. How many objects are in 5 equal rows of 1? Answer: _____

C6. How many objects are in 1 equal rows of 3? Answer: _____

C7. How many objects are in 2 equal rows of 4? Answer: _____

C8. How many objects are in 3 equal rows of 5? Answer: _____

C9. How many objects are in 4 equal rows of 1? Answer: _____

C10. How many objects are in 5 equal rows of 2? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: How many objects are in 3 equal rows of 3?

2. Work with a partner to explain why 16 answers this: How many objects are in 4 equal rows of 4?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: How many objects are in 5 equal rows of 5?

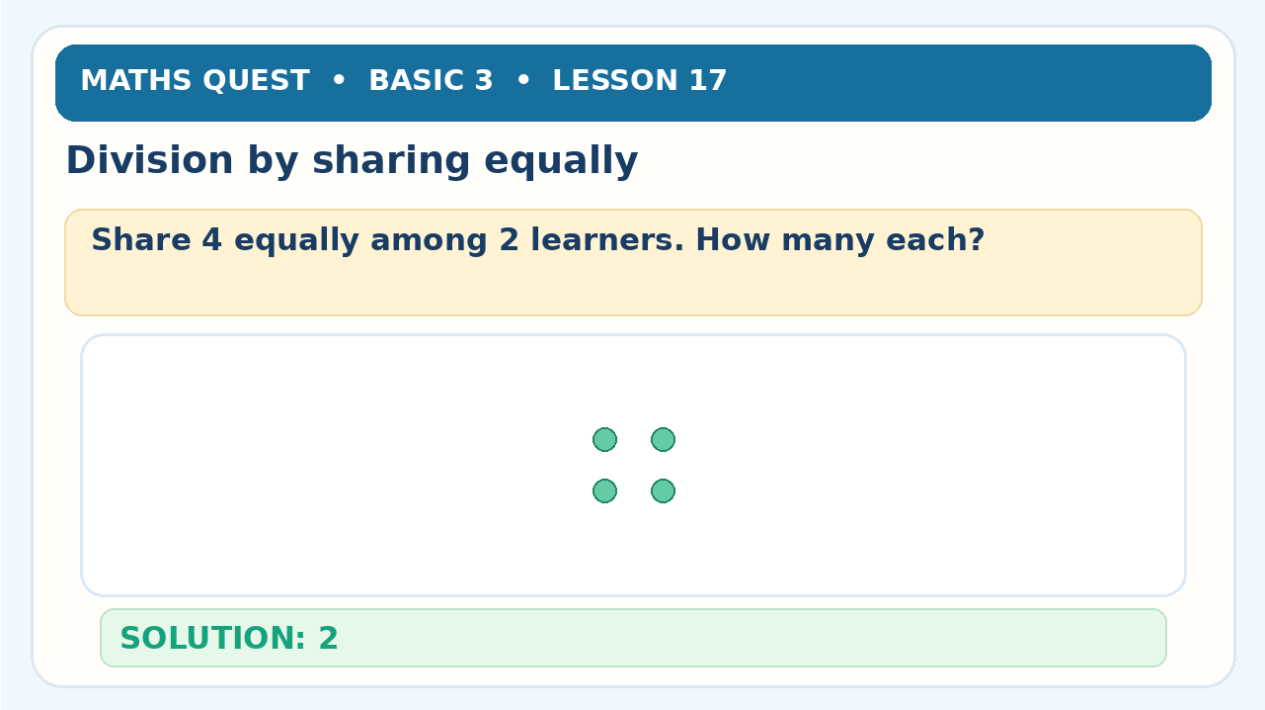
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 17 | Division by sharing equally

YOUR MISSION

Share quantities equally into small groups and interpret the result.



LEARN IT - FOUR IMPORTANT IDEAS

1. Division can mean sharing fairly. If twelve mangoes are shared by three children, each gets four.

2. Draw one counter into each group repeatedly. Check that no group has more than another.

3. Worked example: Share 4 equally among 2 learners. How many each? Answer: 2. 4 ÷ 2 = 2.

4. Check your understanding with another example: Share 6 equally among 3 learners. How many each? The answer is 2. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Share 4 equally among 2 learners. How many each?	2 — 4 ÷ 2 = 2.
Share 6 equally among 3 learners. How many each?	2 — 6 ÷ 3 = 2.
Share 8 equally among 4 learners. How many each?	2 — 8 ÷ 4 = 2.

TRY IT YOURSELF

P1. Share 4 equally among 2 learners. How many each? Answer: _____

P2. Share 6 equally among 3 learners. How many each? Answer: _____

P3. Share 8 equally among 4 learners. How many each? Answer: _____

P4. Share 10 equally among 5 learners. How many each? Answer: _____

P5. Share 6 equally among 2 learners. How many each? Answer: _____

CHALLENGE PRACTICE

C1. Share 9 equally among 3 learners. How many each? Answer: _____

C2. Share 12 equally among 4 learners. How many each? Answer: _____

C3. Share 15 equally among 5 learners. How many each? Answer: _____

C4. Share 8 equally among 2 learners. How many each? Answer: _____

C5. Share 12 equally among 3 learners. How many each? Answer: _____

C6. Share 16 equally among 4 learners. How many each? Answer: _____

C7. Share 20 equally among 5 learners. How many each? Answer: _____

C8. Share 10 equally among 2 learners. How many each? Answer: _____

C9. Share 15 equally among 3 learners. How many each? Answer: _____

C10. Share 20 equally among 4 learners. How many each? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Share 8 equally among 4 learners. How many each?

2. Work with a partner to explain why 2 answers this: Share 10 equally among 5 learners. How many each?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Share 6 equally among 2 learners. How many each?

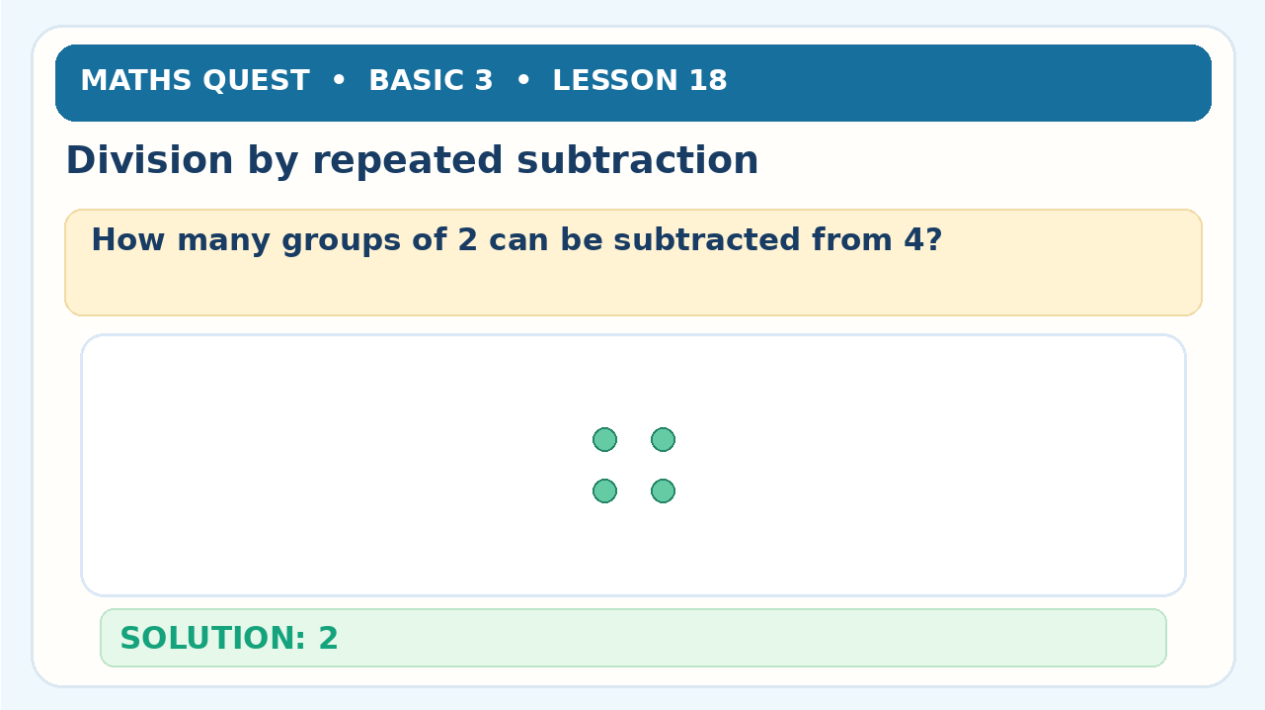
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 18 | Division by repeated subtraction

YOUR MISSION

Use repeated subtraction to find how many equal groups can be removed.



LEARN IT - FOUR IMPORTANT IDEAS

1. Division can also mean finding the number of groups. Start with a total and subtract the same size until zero.

2. For 20 ÷ 5, subtract five four times. The number of subtractions gives the answer four.

3. Worked example: How many groups of 2 can be subtracted from 4? Answer: 2. 4 ÷ 2 = 2.

4. Check your understanding with another example: How many groups of 3 can be subtracted from 6? The answer is 2. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
How many groups of 2 can be subtracted from 4?	2 — 4 ÷ 2 = 2.
How many groups of 3 can be subtracted from 6?	2 — 6 ÷ 3 = 2.
How many groups of 4 can be subtracted from 8?	2 — 8 ÷ 4 = 2.

TRY IT YOURSELF

P1. How many groups of 2 can be subtracted from 4? Answer: _____

P2. How many groups of 3 can be subtracted from 6? Answer: _____

P3. How many groups of 4 can be subtracted from 8? Answer: _____

P4. How many groups of 5 can be subtracted from 10? Answer: _____

P5. How many groups of 2 can be subtracted from 6? Answer: _____

CHALLENGE PRACTICE

C1. How many groups of 3 can be subtracted from 9? Answer: _____

C2. How many groups of 4 can be subtracted from 12? Answer: _____

C3. How many groups of 5 can be subtracted from 15? Answer: _____

C4. How many groups of 2 can be subtracted from 8? Answer: _____

C5. How many groups of 3 can be subtracted from 12? Answer: _____

C6. How many groups of 4 can be subtracted from 16? Answer: _____

C7. How many groups of 5 can be subtracted from 20? Answer: _____

C8. How many groups of 2 can be subtracted from 10? Answer: _____

C9. How many groups of 3 can be subtracted from 15? Answer: _____

C10. How many groups of 4 can be subtracted from 20? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: How many groups of 4 can be subtracted from 8?
2. Work with a partner to explain why 2 answers this: How many groups of 5 can be subtracted from 10?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: How many groups of 2 can be subtracted from 6?

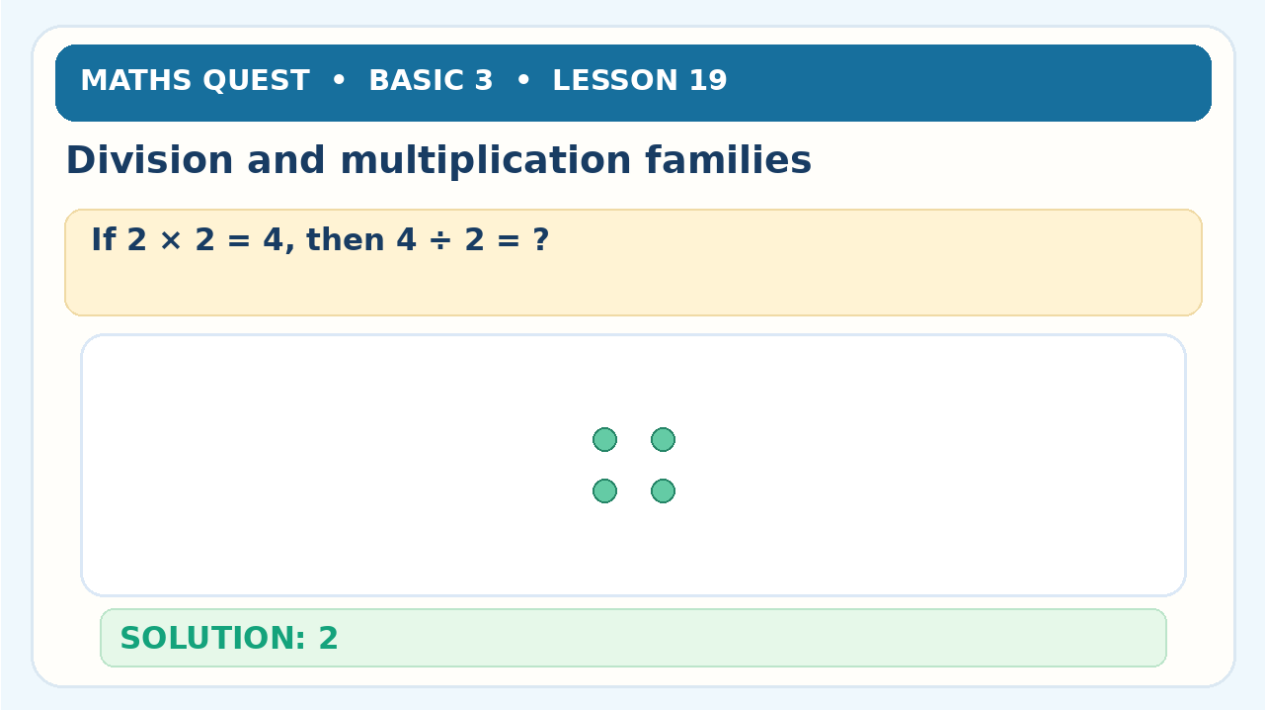
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 19 | Division and multiplication families

YOUR MISSION

Use multiplication to check division and division to check multiplication.



LEARN IT - FOUR IMPORTANT IDEAS

1. Multiplication and division are inverse operations. If 4 × 5 = 20, then 20 ÷ 5 = 4.

2. Arrange counters into a rectangular array and write two multiplication and two division facts from it.

3. Worked example: If 2 × 2 = 4, then 4 ÷ 2 = ? Answer: 2. 4 ÷ 2 = 2.

4. Check your understanding with another example: If 2 × 3 = 6, then 6 ÷ 3 = ? The answer is 2. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
If 2 × 2 = 4, then 4 ÷ 2 = ?	2 — 4 ÷ 2 = 2.
If 2 × 3 = 6, then 6 ÷ 3 = ?	2 — 6 ÷ 3 = 2.
If 2 × 4 = 8, then 8 ÷ 4 = ?	2 — 8 ÷ 4 = 2.

TRY IT YOURSELF

P1. If 2 × 2 = 4, then 4 ÷ 2 = ? Answer: _____

P2. If 2 × 3 = 6, then 6 ÷ 3 = ? Answer: _____

P3. If 2 × 4 = 8, then 8 ÷ 4 = ? Answer: _____

P4. If 2 × 5 = 10, then 10 ÷ 5 = ? Answer: _____

P5. If 3 × 2 = 6, then 6 ÷ 2 = ? Answer: _____

CHALLENGE PRACTICE

C1. If 3 × 3 = 9, then 9 ÷ 3 = ? Answer: _____

C2. If 3 × 4 = 12, then 12 ÷ 4 = ? Answer: _____

C3. If 3 × 5 = 15, then 15 ÷ 5 = ? Answer: _____

C4. If 4 × 2 = 8, then 8 ÷ 2 = ? Answer: _____

C5. If 4 × 3 = 12, then 12 ÷ 3 = ? Answer: _____

C6. If 4 × 4 = 16, then 16 ÷ 4 = ? Answer: _____

C7. If 4 × 5 = 20, then 20 ÷ 5 = ? Answer: _____

C8. If 5 × 2 = 10, then 10 ÷ 2 = ? Answer: _____

C9. If 5 × 3 = 15, then 15 ÷ 3 = ? Answer: _____

C10. If 5 × 4 = 20, then 20 ÷ 4 = ? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: If 2 × 4 = 8, then 8 ÷ 4 = ?

2. Work with a partner to explain why 2 answers this: If 2 × 5 = 10, then 10 ÷ 5 = ?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: If 3 × 2 = 6, then 6 ÷ 2 = ?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 20 | Understand unit fractions

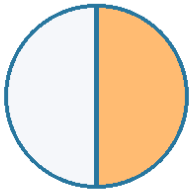
YOUR MISSION

Explain a unit fraction as one equal part of a whole or set.

MATHS QUEST • BASIC 3 • LESSON 20

Understand unit fractions

In pattern card 1, one equal piece of a shape cut into 2 equal parts is which fraction?



1 of 2 equal parts shaded

SOLUTION: 1/2

LEARN IT - FOUR IMPORTANT IDEAS

1. A unit fraction has a numerator of one. The denominator tells how many equal parts form the whole.

2. One quarter means one of four equal parts. All four parts must be the same size.

3. Worked example: In pattern card 1, one equal piece of a shape cut into 2 equal parts is which fraction? Answer: 1/2. A unit fraction is one of 2 equal parts.

4. Check your understanding with another example: In pattern card 2, one equal piece of a shape cut into 3 equal parts is which fraction? The answer is 1/3. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
In pattern card 1, one equal piece of a shape cut into 2 equal parts is which fraction?	1/2 — A unit fraction is one of 2 equal parts.
In pattern card 2, one equal piece of a shape cut into 3 equal parts is which fraction?	1/3 — A unit fraction is one of 3 equal parts.

Try this	How it works
In pattern card 3, one equal piece of a shape cut into 4 equal parts is which fraction?	1/4 — A unit fraction is one of 4 equal parts.

TRY IT YOURSELF

P1. In pattern card 1, one equal piece of a shape cut into 2 equal parts is which fraction? Answer: _____

P2. In pattern card 2, one equal piece of a shape cut into 3 equal parts is which fraction? Answer: _____

P3. In pattern card 3, one equal piece of a shape cut into 4 equal parts is which fraction? Answer: _____

P4. In pattern card 4, one equal piece of a shape cut into 5 equal parts is which fraction? Answer: _____

P5. In pattern card 5, one equal piece of a shape cut into 6 equal parts is which fraction? Answer: _____

CHALLENGE PRACTICE

C1. In pattern card 6, one equal piece of a shape cut into 2 equal parts is which fraction? Answer: _____

C2. In pattern card 7, one equal piece of a shape cut into 3 equal parts is which fraction? Answer: _____

C3. In pattern card 8, one equal piece of a shape cut into 4 equal parts is which fraction? Answer: _____

C4. In pattern card 9, one equal piece of a shape cut into 5 equal parts is which fraction? Answer: _____

C5. In pattern card 10, one equal piece of a shape cut into 6 equal parts is which fraction? Answer: _____

C6. In pattern card 11, one equal piece of a shape cut into 2 equal parts is which fraction? Answer: _____

C7. In pattern card 12, one equal piece of a shape cut into 3 equal parts is which fraction? Answer: _____

C8. In pattern card 13, one equal piece of a shape cut into 4 equal parts is which fraction? Answer: _____

C9. In pattern card 14, one equal piece of a shape cut into 5 equal parts is which fraction? Answer: _____

C10. In pattern card 15, one equal piece of a shape cut into 6 equal parts is which fraction? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: In pattern card 3, one equal piece of a shape cut into 4 equal parts is which fraction?
2. Work with a partner to explain why 1/5 answers this: In pattern card 4, one equal piece of a shape cut into 5 equal parts is which fraction?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: In pattern card 5, one equal piece of a shape cut into 6 equal parts is which fraction?

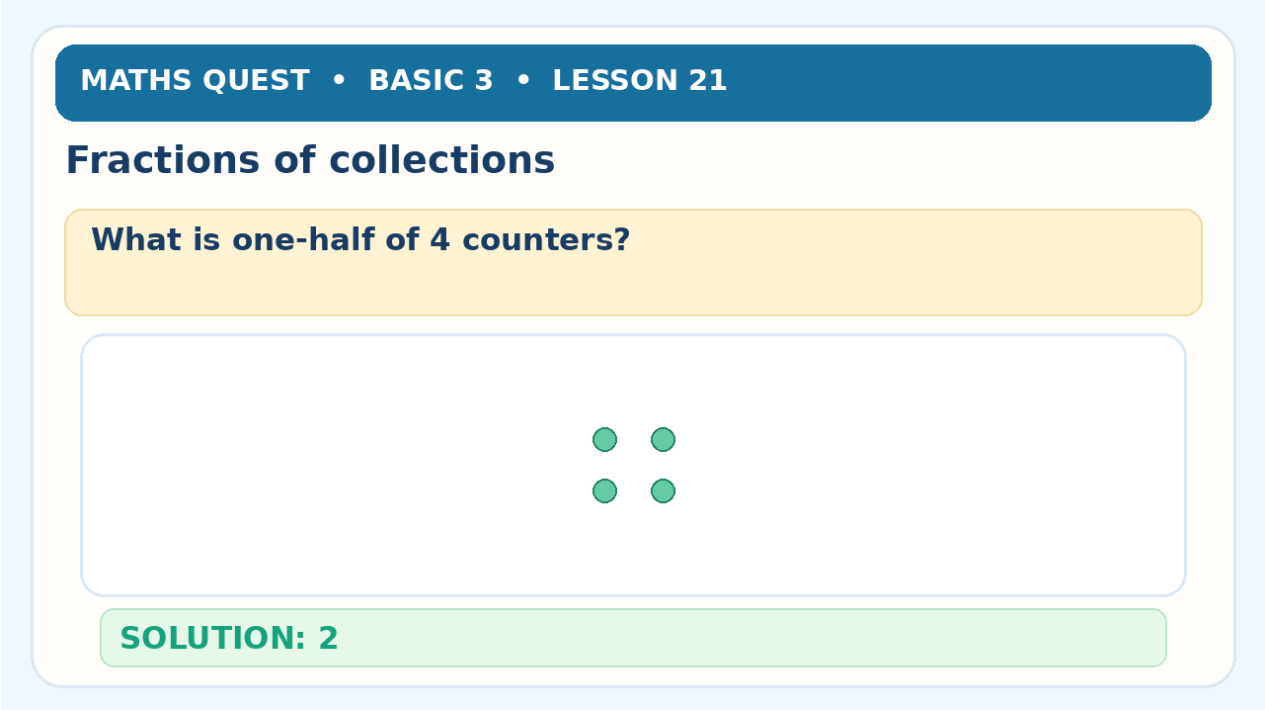
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 21 | Fractions of collections

YOUR MISSION

Find fractional parts of sets and connect them to equal sharing.



LEARN IT - FOUR IMPORTANT IDEAS

1. Fractions can describe part of a group. One half of ten counters is five counters.

2. To find one quarter of twelve, share twelve into four equal groups and choose one group.

3. Worked example: What is one-half of 4 counters? Answer: 2. Share 4 into 2 equal groups to get 2 in each.

4. Check your understanding with another example: What is one-third of 6 counters? The answer is 2. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
What is one-half of 4 counters?	2 — Share 4 into 2 equal groups to get 2 in each.
What is one-third of 6 counters?	2 — Share 6 into 3 equal groups to get 2 in each.
What is one-quarter of 8 counters?	2 — Share 8 into 4 equal groups to get 2 in each.

TRY IT YOURSELF

P1. What is one-half of 4 counters? Answer: _____

P2. What is one-third of 6 counters? Answer: _____

P3. What is one-quarter of 8 counters? Answer: _____

P4. What is one-fifth of 10 counters? Answer: _____

P5. What is one-half of 6 counters? Answer: _____

CHALLENGE PRACTICE

C1. What is one-third of 9 counters? Answer: _____

C2. What is one-quarter of 12 counters? Answer: _____

C3. What is one-fifth of 15 counters? Answer: _____

C4. What is one-half of 8 counters? Answer: _____

C5. What is one-third of 12 counters? Answer: _____

C6. What is one-quarter of 16 counters? Answer: _____

C7. What is one-fifth of 20 counters? Answer: _____

C8. What is one-half of 10 counters? Answer: _____

C9. What is one-third of 15 counters? Answer: _____

C10. What is one-quarter of 20 counters? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: What is one-quarter of 8 counters?

2. Work with a partner to explain why 2 answers this: What is one-fifth of 10 counters?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: What is one-half of 6 counters?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 22 | Compare like-denominator fractions

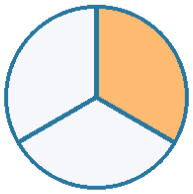
YOUR MISSION

Order unit fractions and fractions with equal denominators.

MATHS QUEST • BASIC 3 • LESSON 22

Compare like-denominator fractions

Fraction card 1: choose a sign: 1/3 __ 2/3



1 of 3 equal parts shaded

SOLUTION: <

LEARN IT - FOUR IMPORTANT IDEAS

1. With the same denominator, the fraction with the greater numerator represents more parts.

2. For unit fractions, bigger denominators make smaller pieces of the same whole. Draw equal-sized wholes before comparing.

3. Worked example: Fraction card 1: choose a sign: 1/3 __ 2/3 Answer: <. With equal denominators, compare the numerators.

4. Check your understanding with another example: Fraction card 2: choose a sign: 2/4 __ 3/4 The answer is <. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Fraction card 1: choose a sign: 1/3 __ 2/3	< — With equal denominators, compare the numerators.
Fraction card 2: choose a sign: 2/4 __ 3/4	< — With equal denominators, compare the numerators.

Try this	How it works
Fraction card 3: choose a sign: 3/5 __ 4/5	< — With equal denominators, compare the numerators.

TRY IT YOURSELF

P1. Fraction card 1: choose a sign: 1/3 __ 2/3 Answer: _____

P2. Fraction card 2: choose a sign: 2/4 __ 3/4 Answer: _____

P3. Fraction card 3: choose a sign: 3/5 __ 4/5 Answer: _____

P4. Fraction card 4: choose a sign: 4/6 __ 5/6 Answer: _____

P5. Fraction card 5: choose a sign: 2/3 __ 3/3 Answer: _____

CHALLENGE PRACTICE

C1. Fraction card 6: choose a sign: 2/4 __ 3/4 Answer: _____

C2. Fraction card 7: choose a sign: 2/5 __ 3/5 Answer: _____

C3. Fraction card 8: choose a sign: 2/6 __ 3/6 Answer: _____

C4. Fraction card 9: choose a sign: 3/3 __ 1/3 Answer: _____

C5. Fraction card 10: choose a sign: 2/4 __ 3/4 Answer: _____

C6. Fraction card 11: choose a sign: 1/5 __ 2/5 Answer: _____

C7. Fraction card 12: choose a sign: 6/6 __ 1/6 Answer: _____

C8. Fraction card 13: choose a sign: 1/3 __ 2/3 Answer: _____

C9. Fraction card 14: choose a sign: 2/4 __ 3/4 Answer: _____

C10. Fraction card 15: choose a sign: 5/5 __ 1/5 Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Fraction card 3: choose a sign: 3/5 __ 4/5

2. Work with a partner to explain why < answers this: Fraction card 4: choose a sign: 4/6 __ 5/6

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Fraction card 5: choose a sign: 2/3 __ 3/3

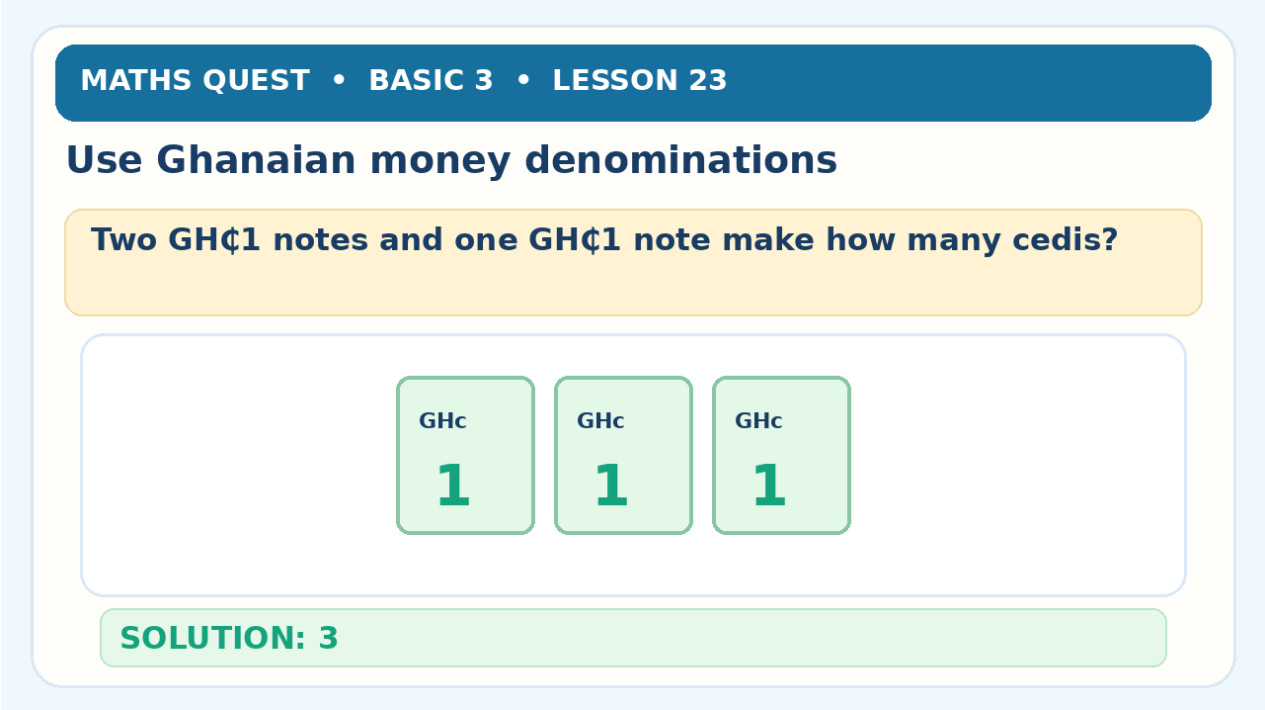
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 23 | Use Ghanaian money denominations

YOUR MISSION

Count notes and coins, make values and calculate simple change.



LEARN IT - FOUR IMPORTANT IDEAS

1. Ghana cedi notes can be combined to make a larger amount. Two ten-cedi notes and one five-cedi note total 25 cedis.

2. Different combinations may have the same total. When buying an item, change equals money paid minus the price.

3. Worked example: Two GH¢1 notes and one GH¢1 note make how many cedis? Answer: 3. Add 1 + 1 + 1.

4. Check your understanding with another example: Two GH¢2 notes and one GH¢2 note make how many cedis? The answer is 6. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Two GH¢1 notes and one GH¢1 note make how many cedis?	3 — Add 1 + 1 + 1.

Try this	How it works
Two GH¢2 notes and one GH¢2 note make how many cedis?	6 — Add 2 + 2 + 2.
Two GH¢5 notes and one GH¢5 note make how many cedis?	15 — Add 5 + 5 + 5.

TRY IT YOURSELF

P1. Two GH¢1 notes and one GH¢1 note make how many cedis? Answer: _____

P2. Two GH¢2 notes and one GH¢2 note make how many cedis? Answer: _____

P3. Two GH¢5 notes and one GH¢5 note make how many cedis? Answer: _____

P4. Two GH¢10 notes and one GH¢10 note make how many cedis? Answer: _____

P5. Two GH¢20 notes and one GH¢1 note make how many cedis? Answer: _____

CHALLENGE PRACTICE

C1. Two GH¢50 notes and one GH¢2 note make how many cedis? Answer: _____

C2. Two GH¢1 notes and one GH¢5 note make how many cedis? Answer: _____

C3. Two GH¢2 notes and one GH¢10 note make how many cedis? Answer: _____

C4. Two GH¢5 notes and one GH¢1 note make how many cedis? Answer: _____

C5. Two GH¢10 notes and one GH¢2 note make how many cedis? Answer: _____

C6. Two GH¢20 notes and one GH¢5 note make how many cedis? Answer: _____

C7. Two GH¢50 notes and one GH¢10 note make how many cedis? Answer: _____

C8. Practice card 13: Two GH¢1 notes and one GH¢1 note make how many cedis? Answer: _____

C9. Practice card 14: Two GH¢2 notes and one GH¢2 note make how many cedis? Answer: _____

C10. Practice card 15: Two GH¢5 notes and one GH¢5 note make how many cedis? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Two GH¢5 notes and one GH¢5 note make how many cedis?
2. Work with a partner to explain why 30 answers this: Two GH¢10 notes and one GH¢10 note make how many cedis?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Two GH¢20 notes and one GH¢1 note make how many cedis?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 24 | Investigate growing and shrinking patterns

YOUR MISSION

Extend and describe increasing and decreasing number patterns.



LEARN IT - FOUR IMPORTANT IDEAS

1. Find the difference between neighbouring terms. In 12, 16, 20, 24, the rule is add four.

2. A decreasing pattern follows a subtraction rule, such as 50, 45, 40, 35. Test your rule against every term.

3. Worked example: Continue the pattern: 40, 42, 44, __ Answer: 46. The repeated difference is 2.

4. Check your understanding with another example: Continue the pattern: 43, 46, 49, __ The answer is 52. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Continue the pattern: 40, 42, 44, __	46 — The repeated difference is 2.
Continue the pattern: 43, 46, 49, __	52 — The repeated difference is 3.
Continue the pattern: 46, 50, 54, __	58 — The repeated difference is 4.

TRY IT YOURSELF

P1. Continue the pattern: 40, 42, 44, __ Answer: _____

P2. Continue the pattern: 43, 46, 49, __ Answer: _____

P3. Continue the pattern: 46, 50, 54, __ Answer: _____

P4. Continue the pattern: 49, 54, 59, __ Answer: _____

P5. Continue the pattern: 52, 50, 48, __ Answer: _____

CHALLENGE PRACTICE

C1. Continue the pattern: 55, 52, 49, __ Answer: _____

C2. Continue the pattern: 58, 60, 62, __ Answer: _____

C3. Continue the pattern: 61, 64, 67, __ Answer: _____

C4. Continue the pattern: 64, 68, 72, __ Answer: _____

C5. Continue the pattern: 67, 72, 77, __ Answer: _____

C6. Continue the pattern: 70, 68, 66, __ Answer: _____

C7. Continue the pattern: 73, 70, 67, __ Answer: _____

C8. Continue the pattern: 76, 78, 80, __ Answer: _____

C9. Continue the pattern: 79, 82, 85, __ Answer: _____

C10. Continue the pattern: 82, 86, 90, __ Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Continue the pattern: 46, 50, 54, __

2. Work with a partner to explain why 64 answers this: Continue the pattern: 49, 54, 59, __

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Continue the pattern: 52, 50, 48, __

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 25 | Discover faces, edges and vertices

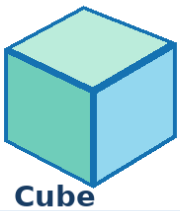
YOUR MISSION

Describe familiar 3D objects by their faces, edges and corners.

MATHS QUEST • BASIC 3 • LESSON 25

Discover faces, edges and vertices

Model card 1: how many faces does a cube have?



Cube

SOLUTION: 6

LEARN IT - FOUR IMPORTANT IDEAS

1. A cube has six square faces, twelve edges and eight vertices. An edge is where faces meet.

2. A cylinder has two flat circular faces and a curved surface. Compare objects by shape as well as size.

3. Worked example: Model card 1: how many faces does a cube have? Answer: 6. A cube has 6 faces.

4. Check your understanding with another example: Model card 2: how many faces does a cuboid have? The answer is 6. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Model card 1: how many faces does a cube have?	6 — A cube has 6 faces.
Model card 2: how many faces does a cuboid have?	6 — A cuboid has 6 faces.

Try this	How it works
Model card 3: how many faces does a square pyramid have?	5 — A square pyramid has 5 faces.

TRY IT YOURSELF

P1. Model card 1: how many faces does a cube have? Answer: _____

P2. Model card 2: how many faces does a cuboid have? Answer: _____

P3. Model card 3: how many faces does a square pyramid have? Answer: _____

P4. Model card 4: how many flat circular faces does a cylinder have? Answer: _____

P5. Model card 5: how many flat faces does a sphere have? Answer: _____

CHALLENGE PRACTICE

C1. Model card 6: how many flat circular face does a cone have? Answer: _____

C2. Model card 7: how many faces does a cube have? Answer: _____

C3. Model card 8: how many faces does a cuboid have? Answer: _____

C4. Model card 9: how many faces does a square pyramid have? Answer: _____

C5. Model card 10: how many flat circular faces does a cylinder have? Answer: _____

C6. Model card 11: how many flat faces does a sphere have? Answer: _____

C7. Model card 12: how many flat circular face does a cone have? Answer: _____

C8. Model card 13: how many faces does a cube have? Answer: _____

C9. Model card 14: how many faces does a cuboid have? Answer: _____

C10. Model card 15: how many faces does a square pyramid have? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Model card 3: how many faces does a square pyramid have?
2. Work with a partner to explain why 2 answers this: Model card 4: how many flat circular faces does a cylinder have?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Model card 5: how many flat faces does a sphere have?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 26 | Draw and recognise angles

YOUR MISSION

Identify angles formed where two rays or line segments meet.

MATHS QUEST • BASIC 3 • LESSON 26

Draw and recognise angles

An angle opens 20 degrees. Is it acute, right or obtuse?



20 degrees

SOLUTION: acute

LEARN IT - FOUR IMPORTANT IDEAS

1. An angle is the opening between two rays meeting at a vertex. The amount of turning shows its size.

2. You can draw angles by drawing two arms from the same point. Compare the openings, not the arm lengths.

3. Worked example: An angle opens 20 degrees. Is it acute, right or obtuse? Answer: acute. Acute is less than 90°, right is 90°, and obtuse is between 90° and 180°.

4. Check your understanding with another example: An angle opens 30 degrees. Is it acute, right or obtuse? The answer is acute. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
An angle opens 20 degrees. Is it acute, right or obtuse?	acute — Acute is less than 90°, right is 90°, and obtuse is between 90° and 180°.
An angle opens 30 degrees. Is it acute, right or obtuse?	acute — Acute is less than 90°, right is 90°, and obtuse is between 90° and 180°.

Try this	How it works
An angle opens 35 degrees. Is it acute, right or obtuse?	acute — Acute is less than 90°, right is 90°, and obtuse is between 90° and 180°.

TRY IT YOURSELF

P1. An angle opens 20 degrees. Is it acute, right or obtuse? Answer: _____

P2. An angle opens 30 degrees. Is it acute, right or obtuse? Answer: _____

P3. An angle opens 35 degrees. Is it acute, right or obtuse? Answer: _____

P4. An angle opens 40 degrees. Is it acute, right or obtuse? Answer: _____

P5. An angle opens 45 degrees. Is it acute, right or obtuse? Answer: _____

CHALLENGE PRACTICE

C1. An angle opens 50 degrees. Is it acute, right or obtuse? Answer: _____

C2. An angle opens 60 degrees. Is it acute, right or obtuse? Answer: _____

C3. An angle opens 65 degrees. Is it acute, right or obtuse? Answer: _____

C4. An angle opens 75 degrees. Is it acute, right or obtuse? Answer: _____

C5. An angle opens 80 degrees. Is it acute, right or obtuse? Answer: _____

C6. An angle opens 90 degrees. Is it acute, right or obtuse? Answer: _____

C7. An angle opens 100 degrees. Is it acute, right or obtuse? Answer: _____

C8. An angle opens 110 degrees. Is it acute, right or obtuse? Answer: _____

C9. An angle opens 125 degrees. Is it acute, right or obtuse? Answer: _____

C10. An angle opens 145 degrees. Is it acute, right or obtuse? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: An angle opens 35 degrees. Is it acute, right or obtuse?
2. Work with a partner to explain why acute answers this: An angle opens 40 degrees. Is it acute, right or obtuse?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: An angle opens 45 degrees. Is it acute, right or obtuse?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 27 | Compare angles with a square corner

YOUR MISSION

Use a folded square corner to identify smaller, equal and larger angles.

MATHS QUEST • BASIC 3 • LESSON 27

Compare angles with a square corner

Compared to a square corner, is a 25-degree angle smaller, equal or larger?



25 degrees

SOLUTION: smaller

LEARN IT - FOUR IMPORTANT IDEAS

1. The corner of a square is a right angle. Use it as a reference when comparing other angles.

2. An angle smaller than a right angle is acute; one larger than a right angle and smaller than a straight angle is obtuse.

3. Worked example: Compared to a square corner, is a 25-degree angle smaller, equal or larger? Answer: smaller. A square corner is 90 degrees.

4. Check your understanding with another example: Compared to a square corner, is a 30-degree angle smaller, equal or larger? The answer is smaller. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Compared to a square corner, is a 25-degree angle smaller, equal or larger?	smaller — A square corner is 90 degrees.
Compared to a square corner, is a 30-degree angle smaller, equal or larger?	smaller — A square corner is 90 degrees.

Try this	How it works
Compared to a square corner, is a 40-degree angle smaller, equal or larger?	smaller — A square corner is 90 degrees.

TRY IT YOURSELF

P1. Compared to a square corner, is a 25-degree angle smaller, equal or larger? Answer: _____

P2. Compared to a square corner, is a 30-degree angle smaller, equal or larger? Answer: _____

P3. Compared to a square corner, is a 40-degree angle smaller, equal or larger? Answer: _____

P4. Compared to a square corner, is a 50-degree angle smaller, equal or larger? Answer: _____

P5. Compared to a square corner, is a 60-degree angle smaller, equal or larger? Answer: _____

CHALLENGE PRACTICE

C1. Compared to a square corner, is a 70-degree angle smaller, equal or larger? Answer: _____

C2. Compared to a square corner, is a 80-degree angle smaller, equal or larger? Answer: _____

C3. Compared to a square corner, is a 90-degree angle smaller, equal or larger? Answer: _____

C4. Compared to a square corner, is a 100-degree angle smaller, equal or larger? Answer: _____

C5. Compared to a square corner, is a 105-degree angle smaller, equal or larger? Answer: _____

C6. Compared to a square corner, is a 115-degree angle smaller, equal or larger? Answer: _____

C7. Compared to a square corner, is a 125-degree angle smaller, equal or larger? Answer: _____

C8. Compared to a square corner, is a 135-degree angle smaller, equal or larger? Answer: _____

C9. Compared to a square corner, is a 145-degree angle smaller, equal or larger? Answer: _____

C10. Compared to a square corner, is a 155-degree angle smaller, equal or larger? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Compared to a square corner, is a 40-degree angle smaller, equal or larger?

2. Work with a partner to explain why smaller answers this: Compared to a square corner, is a 50-degree angle smaller, equal or larger?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Compared to a square corner, is a 60-degree angle smaller, equal or larger?

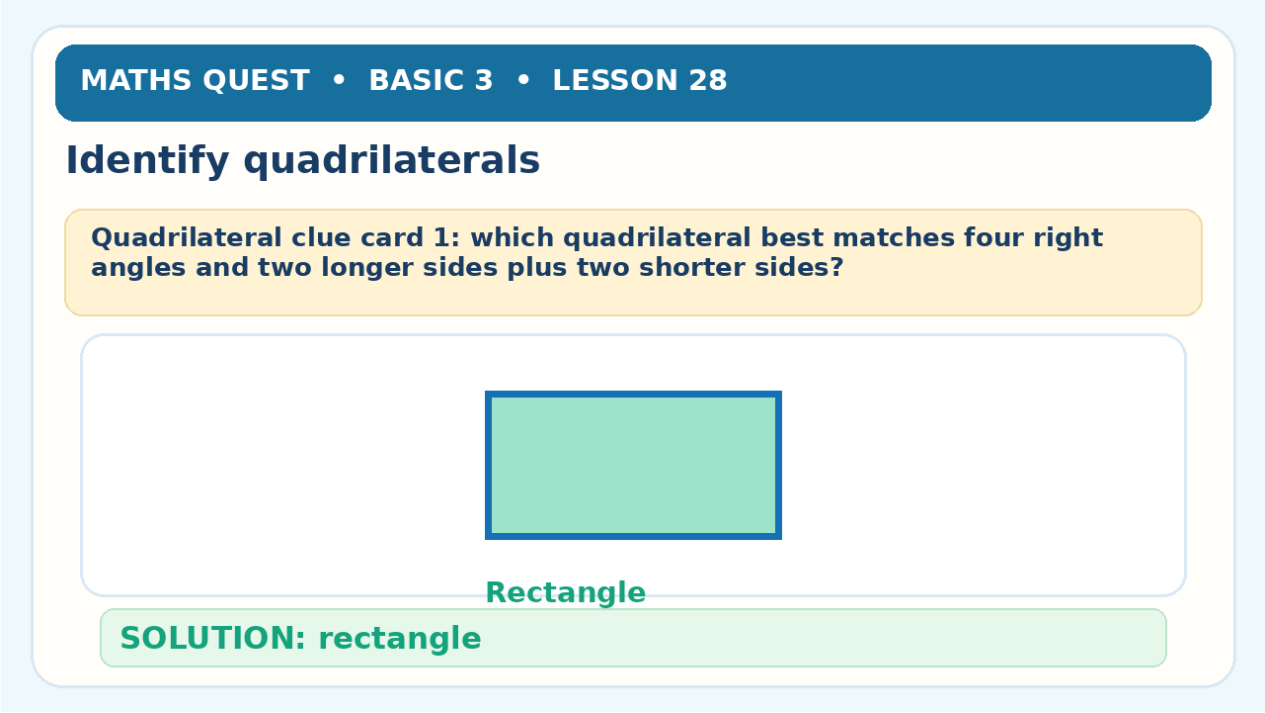
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 28 | Identify quadrilaterals

YOUR MISSION

Recognise rectangles, squares, parallelograms, rhombuses and trapezoids.



LEARN IT - FOUR IMPORTANT IDEAS

1. A quadrilateral has four straight sides. A rectangle has four right angles. A square has four equal sides and four right angles.

2. A rhombus has four equal sides; a parallelogram has two pairs of parallel sides. Compare pictures and physical cut-outs.

3. Worked example: Quadrilateral clue card 1: which quadrilateral best matches four right angles and two longer sides plus two shorter sides? Answer: rectangle. Rectangle matches these attributes. Some categories can overlap; use the intended most specific name.

4. Check your understanding with another example: Quadrilateral clue card 2: which quadrilateral best matches four equal sides and four right angles? The answer is square. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Quadrilateral clue card 1: which quadrilateral best matches four right angles and two longer sides plus two shorter sides?	rectangle — Rectangle matches these attributes. Some categories can overlap; use the intended most specific name.

Try this	How it works
Quadrilateral clue card 2: which quadrilateral best matches four equal sides and four right angles?	square — Square matches these attributes. Some categories can overlap; use the intended most specific name.
Quadrilateral clue card 3: which quadrilateral best matches four equal sides and no right angles?	rhombus — Rhombus matches these attributes. Some categories can overlap; use the intended most specific name.

TRY IT YOURSELF

P1. Quadrilateral clue card 1: which quadrilateral best matches four right angles and two longer sides plus two shorter sides? Answer: _____

P2. Quadrilateral clue card 2: which quadrilateral best matches four equal sides and four right angles? Answer: _____

P3. Quadrilateral clue card 3: which quadrilateral best matches four equal sides and no right angles? Answer: _____

P4. Quadrilateral clue card 4: which quadrilateral best matches two pairs of parallel sides, no right angles and adjacent sides of different lengths? Answer: _____

P5. Quadrilateral clue card 5: which quadrilateral best matches exactly one pair of parallel sides? Answer: _____

CHALLENGE PRACTICE

C1. Quadrilateral clue card 6: which quadrilateral best matches four right angles and two longer sides plus two shorter sides? Answer: _____

C2. Quadrilateral clue card 7: which quadrilateral best matches four equal sides and four right angles? Answer: _____

C3. Quadrilateral clue card 8: which quadrilateral best matches four equal sides and no right angles? Answer: _____

C4. Quadrilateral clue card 9: which quadrilateral best matches two pairs of parallel sides, no right angles and adjacent sides of different lengths? Answer: _____

C5. Quadrilateral clue card 10: which quadrilateral best matches exactly one pair of parallel sides? Answer: _____

C6. Quadrilateral clue card 11: which quadrilateral best matches four right angles and two longer sides plus two shorter sides? Answer: _____

C7. Quadrilateral clue card 12: which quadrilateral best matches four equal sides and four right angles? Answer: _____

C8. Quadrilateral clue card 13: which quadrilateral best matches four equal sides and no right angles? Answer: _____

C9. Quadrilateral clue card 14: which quadrilateral best matches two pairs of parallel sides, no right angles and adjacent sides of different lengths? Answer: _____

C10. Quadrilateral clue card 15: which quadrilateral best matches exactly one pair of parallel sides? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Quadrilateral clue card 3: which quadrilateral best matches four equal sides and no right angles?

2. Work with a partner to explain why parallelogram answers this: Quadrilateral clue card 4: which quadrilateral best matches two pairs of parallel sides, no right angles and adjacent sides of different lengths?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Quadrilateral clue card 5: which quadrilateral best matches exactly one pair of parallel sides?

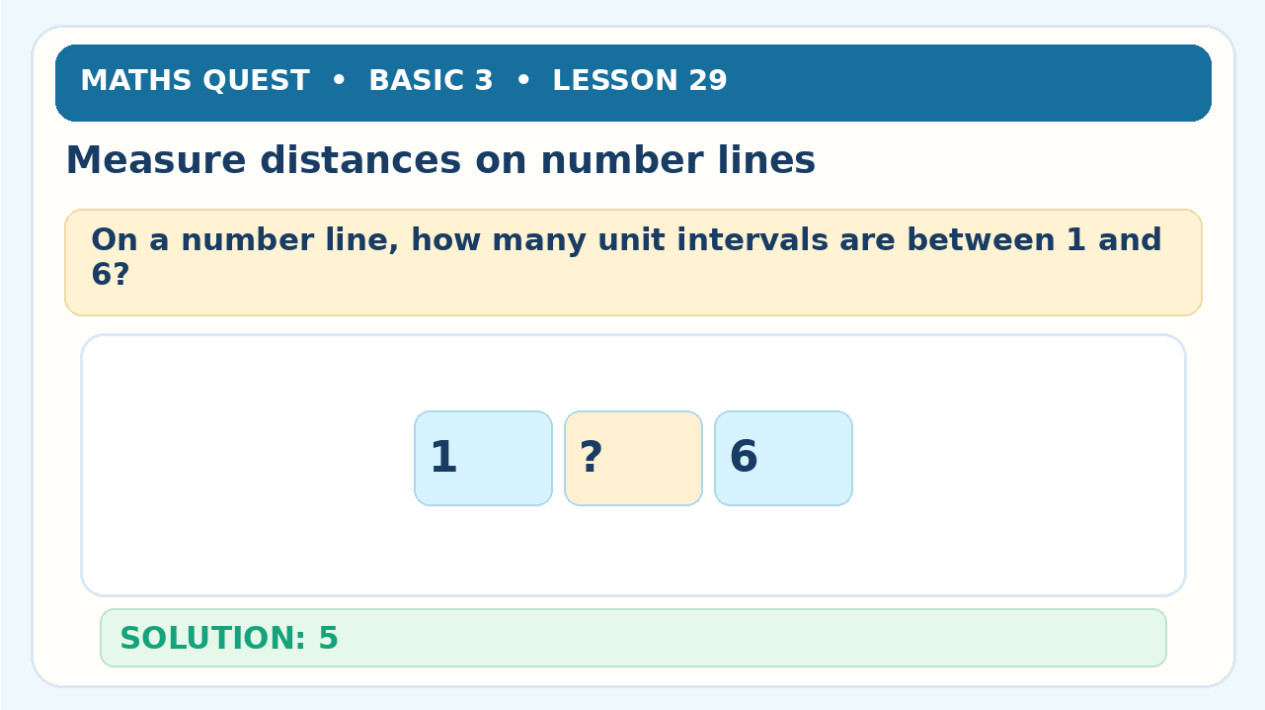
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 29 | Measure distances on number lines

YOUR MISSION

Show a fixed length as a distance between number-line positions.



LEARN IT - FOUR IMPORTANT IDEAS

1. Distance is the number of equal intervals between two points. The distance from 3 to 9 is six units.

2. Sliding an object along a number line does not change its length. Draw both placements to show the same distance.

3. Worked example: On a number line, how many unit intervals are between 1 and 6? Answer: 5. Subtract the starting point from the end point.

4. Check your understanding with another example: On a number line, how many unit intervals are between 3 and 9? The answer is 6. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
On a number line, how many unit intervals are between 1 and 6?	5 — Subtract the starting point from the end point.

Try this	How it works
On a number line, how many unit intervals are between 3 and 9?	6 — Subtract the starting point from the end point.
On a number line, how many unit intervals are between 5 and 12?	7 — Subtract the starting point from the end point.

TRY IT YOURSELF

P1. On a number line, how many unit intervals are between 1 and 6? Answer: _____

P2. On a number line, how many unit intervals are between 3 and 9? Answer: _____

P3. On a number line, how many unit intervals are between 5 and 12? Answer: _____

P4. On a number line, how many unit intervals are between 7 and 15? Answer: _____

P5. On a number line, how many unit intervals are between 9 and 18? Answer: _____

CHALLENGE PRACTICE

C1. On a number line, how many unit intervals are between 11 and 21? Answer: _____

C2. On a number line, how many unit intervals are between 13 and 24? Answer: _____

C3. On a number line, how many unit intervals are between 15 and 20? Answer: _____

C4. On a number line, how many unit intervals are between 17 and 23? Answer: _____

C5. On a number line, how many unit intervals are between 19 and 26? Answer: _____

C6. On a number line, how many unit intervals are between 21 and 29? Answer: _____

C7. On a number line, how many unit intervals are between 23 and 32? Answer: _____

C8. On a number line, how many unit intervals are between 25 and 35? Answer: _____

C9. On a number line, how many unit intervals are between 27 and 38? Answer: _____

C10. On a number line, how many unit intervals are between 29 and 34? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: On a number line, how many unit intervals are between 5 and 12?
2. Work with a partner to explain why 8 answers this: On a number line, how many unit intervals are between 7 and 15?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: On a number line, how many unit intervals are between 9 and 18?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 30 | Metres and centimetres

YOUR MISSION

Describe and use 1 metre = 100 centimetres for length measurement.

MATHS QUEST • BASIC 3 • LESSON 30

Metres and centimetres

How many centimetres are in 1 metres?



SOLUTION: 100

LEARN IT - FOUR IMPORTANT IDEAS

1. A centimetre is a small unit of length. A metre is 100 centimetres.

2. Choose metres for a classroom wall and centimetres for the width of a book. Begin ruler measurement at the zero mark.

3. Worked example: How many centimetres are in 1 metres? Answer: 100. Multiply metres by 100 because 1 m = 100 cm.

4. Check your understanding with another example: How many centimetres are in 2 metres? The answer is 200. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
How many centimetres are in 1 metres?	100 — Multiply metres by 100 because 1 m = 100 cm.
How many centimetres are in 2 metres?	200 — Multiply metres by 100 because 1 m = 100 cm.

Try this	How it works
How many centimetres are in 3 metres?	300 — Multiply metres by 100 because 1 m = 100 cm.

TRY IT YOURSELF

P1. How many centimetres are in 1 metres? Answer: _____

P2. How many centimetres are in 2 metres? Answer: _____

P3. How many centimetres are in 3 metres? Answer: _____

P4. How many centimetres are in 4 metres? Answer: _____

P5. How many centimetres are in 5 metres? Answer: _____

CHALLENGE PRACTICE

C1. How many centimetres are in 6 metres? Answer: _____

C2. How many centimetres are in 7 metres? Answer: _____

C3. How many centimetres are in 8 metres? Answer: _____

C4. How many centimetres are in 9 metres? Answer: _____

C5. How many centimetres are in 10 metres? Answer: _____

C6. How many centimetres are in 11 metres? Answer: _____

C7. How many centimetres are in 12 metres? Answer: _____

C8. How many centimetres are in 13 metres? Answer: _____

C9. How many centimetres are in 14 metres? Answer: _____

C10. How many centimetres are in 15 metres? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: How many centimetres are in 3 metres?

2. Work with a partner to explain why 400 answers this: How many centimetres are in 4 metres?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: How many centimetres are in 5 metres?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 31 | Choose suitable length units

YOUR MISSION

Choose cm or m based on the size of an everyday object.

MATHS QUEST • BASIC 3 • LESSON 31

Choose suitable length units

Item 1: which unit is sensible for measuring the length of a pencil?



SOLUTION: cm

LEARN IT - FOUR IMPORTANT IDEAS

1. A pencil is usually measured in centimetres; a door can be measured in metres.

2. Choosing a sensible unit makes an answer easy to understand. Explain your choice instead of guessing.

3. Worked example: Item 1: which unit is sensible for measuring the length of a pencil? Answer: cm. Use centimetres for shorter lengths and metres for longer lengths.

4. Check your understanding with another example: Item 2: which unit is sensible for measuring the length of a door? The answer is m. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Item 1: which unit is sensible for measuring the length of a pencil?	cm — Use centimetres for shorter lengths and metres for longer lengths.
Item 2: which unit is sensible for measuring the length of a door?	m — Use centimetres for shorter lengths and metres for longer lengths.

Try this	How it works
Item 3: which unit is sensible for measuring the length of a textbook?	cm — Use centimetres for shorter lengths and metres for longer lengths.

TRY IT YOURSELF

P1. Item 1: which unit is sensible for measuring the length of a pencil? Answer: _____

P2. Item 2: which unit is sensible for measuring the length of a door? Answer: _____

P3. Item 3: which unit is sensible for measuring the length of a textbook? Answer: _____

P4. Item 4: which unit is sensible for measuring the length of a football pitch? Answer: _____

P5. Item 5: which unit is sensible for measuring the length of a finger? Answer: _____

CHALLENGE PRACTICE

C1. Item 6: which unit is sensible for measuring the length of a classroom wall? Answer: _____

C2. Item 7: which unit is sensible for measuring the length of a shoelace? Answer: _____

C3. Item 8: which unit is sensible for measuring the length of a schoolyard? Answer: _____

C4. Item 9: which unit is sensible for measuring the length of a pencil? Answer: _____

C5. Item 10: which unit is sensible for measuring the length of a door? Answer: _____

C6. Item 11: which unit is sensible for measuring the length of a textbook? Answer: _____

C7. Item 12: which unit is sensible for measuring the length of a football pitch? Answer: _____

C8. Item 13: which unit is sensible for measuring the length of a finger? Answer: _____

C9. Item 14: which unit is sensible for measuring the length of a classroom wall? Answer: _____

C10. Item 15: which unit is sensible for measuring the length of a shoelace? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Item 3: which unit is sensible for measuring the length of a textbook?
2. Work with a partner to explain why m answers this: Item 4: which unit is sensible for measuring the length of a football pitch?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Item 5: which unit is sensible for measuring the length of a finger?

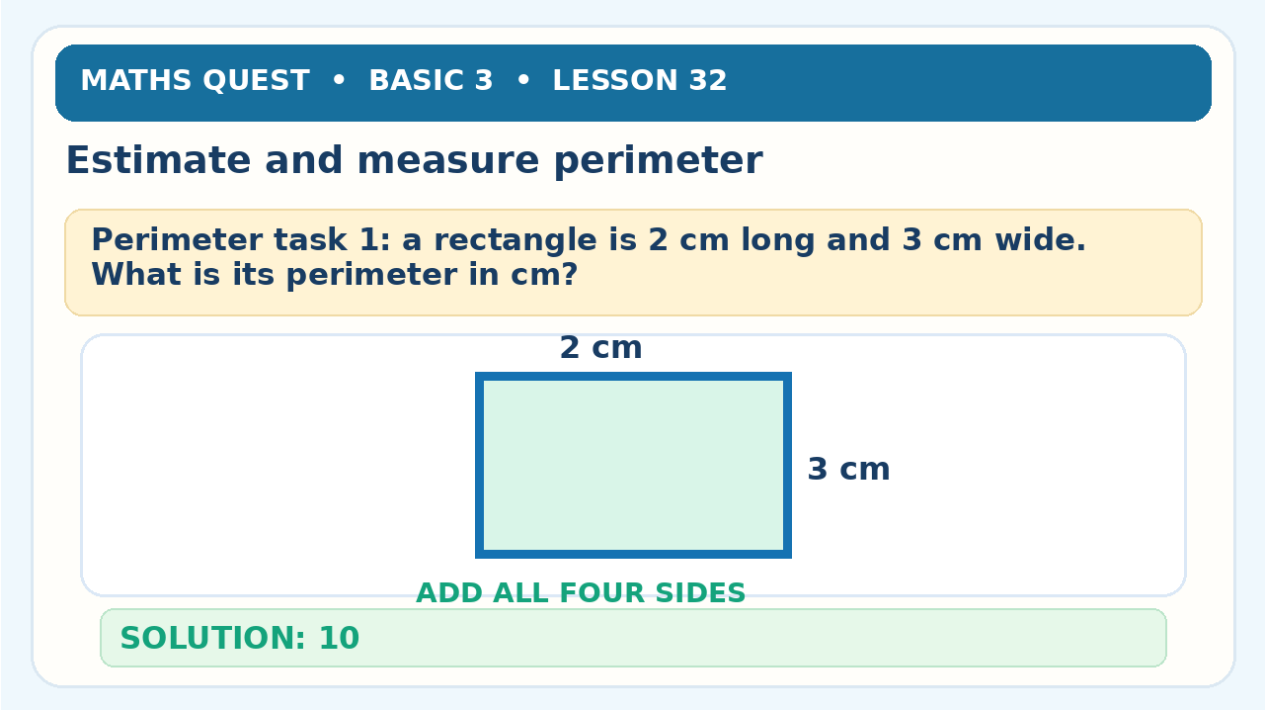
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 32 | Estimate and measure perimeter

YOUR MISSION

Find perimeter by adding the lengths of the sides of a closed shape.



LEARN IT - FOUR IMPORTANT IDEAS

1. Perimeter is the total distance around a flat shape. Add every outside side exactly once.

2. A rectangle 5 cm long and 3 cm wide has a perimeter of 5 + 3 + 5 + 3 = 16 cm.

3. Worked example: Perimeter task 1: a rectangle is 2 cm long and 3 cm wide. What is its perimeter in cm? Answer: 10. Add every side: 2+3+2+3=10 cm.

4. Check your understanding with another example: Perimeter task 2: a rectangle is 3 cm long and 5 cm wide. What is its perimeter in cm? The answer is 16. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Perimeter task 1: a rectangle is 2 cm long and 3 cm wide. What is its perimeter in cm?	10 — Add every side: 2+3+2+3=10 cm.
Perimeter task 2: a rectangle is 3 cm long and 5 cm wide. What is its perimeter in cm?	16 — Add every side: 3+5+3+5=16 cm.

Try this	How it works
Perimeter task 3: a rectangle is 4 cm long and 7 cm wide. What is its perimeter in cm?	22 — Add every side: 4+7+4+7=22 cm.

TRY IT YOURSELF

P1. Perimeter task 1: a rectangle is 2 cm long and 3 cm wide. What is its perimeter in cm? Answer: _____

P2. Perimeter task 2: a rectangle is 3 cm long and 5 cm wide. What is its perimeter in cm? Answer: _____

P3. Perimeter task 3: a rectangle is 4 cm long and 7 cm wide. What is its perimeter in cm? Answer: _____

P4. Perimeter task 4: a rectangle is 5 cm long and 9 cm wide. What is its perimeter in cm? Answer: _____

P5. Perimeter task 5: a rectangle is 6 cm long and 4 cm wide. What is its perimeter in cm? Answer: _____

CHALLENGE PRACTICE

C1. Perimeter task 6: a rectangle is 7 cm long and 6 cm wide. What is its perimeter in cm? Answer: _____

C2. Perimeter task 7: a rectangle is 8 cm long and 8 cm wide. What is its perimeter in cm? Answer: _____

C3. Perimeter task 8: a rectangle is 9 cm long and 3 cm wide. What is its perimeter in cm? Answer: _____

C4. Perimeter task 9: a rectangle is 2 cm long and 5 cm wide. What is its perimeter in cm? Answer: _____

C5. Perimeter task 10: a rectangle is 3 cm long and 7 cm wide. What is its perimeter in cm? Answer: _____

C6. Perimeter task 11: a rectangle is 4 cm long and 9 cm wide. What is its perimeter in cm? Answer: _____

C7. Perimeter task 12: a rectangle is 5 cm long and 4 cm wide. What is its perimeter in cm? Answer: _____

C8. Perimeter task 13: a rectangle is 6 cm long and 6 cm wide. What is its perimeter in cm? Answer: _____

C9. Perimeter task 14: a rectangle is 7 cm long and 8 cm wide. What is its perimeter in cm? Answer: _____

C10. Perimeter task 15: a rectangle is 8 cm long and 3 cm wide. What is its perimeter in cm? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Perimeter task 3: a rectangle is 4 cm long and 7 cm wide. What is its perimeter in cm?
2. Work with a partner to explain why 28 answers this: Perimeter task 4: a rectangle is 5 cm long and 9 cm wide. What is its perimeter in cm?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Perimeter task 5: a rectangle is 6 cm long and 4 cm wide. What is its perimeter in cm?

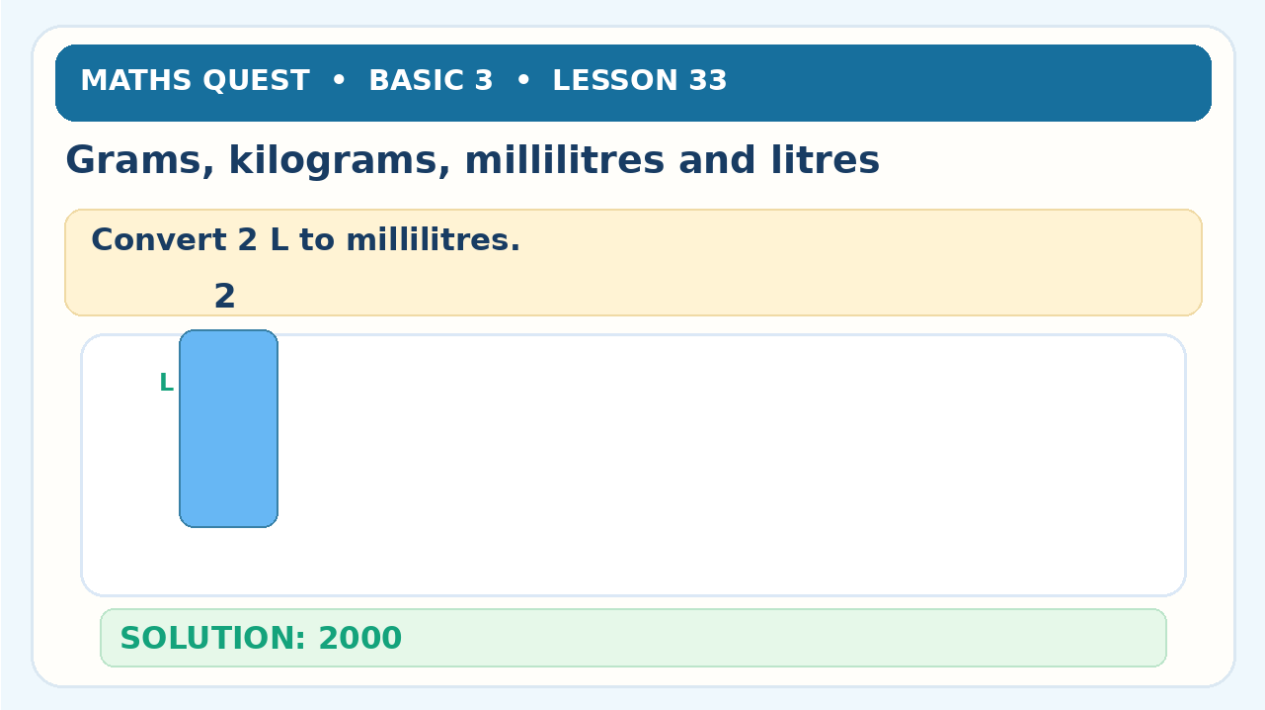
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 33 | Grams, kilograms, millilitres and litres

YOUR MISSION

Relate 1 kg to 1,000 g and 1 L to 1,000 mL.



LEARN IT - FOUR IMPORTANT IDEAS

1. Mass measures how heavy an object is. One kilogram is equal to one thousand grams.

2. Capacity measures liquid a container can hold. One litre is equal to one thousand millilitres.

3. Worked example: Convert 2 L to millilitres. Answer: 2000. One kilogram has 1,000 grams and one litre has 1,000 millilitres.

4. Check your understanding with another example: Convert 5 kg to grams. The answer is 5000. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Convert 2 L to millilitres.	2000 — One kilogram has 1,000 grams and one litre has 1,000 millilitres.
Convert 5 kg to grams.	5000 — One kilogram has 1,000 grams and one litre has 1,000 millilitres.

Try this	How it works
Convert 3 L to millilitres.	3000 — One kilogram has 1,000 grams and one litre has 1,000 millilitres.

TRY IT YOURSELF

P1. Convert 2 L to millilitres. Answer: _____

P2. Convert 5 kg to grams. Answer: _____

P3. Convert 3 L to millilitres. Answer: _____

P4. Convert 4 kg to grams. Answer: _____

P5. Convert 6 L to millilitres. Answer: _____

CHALLENGE PRACTICE

C1. Convert 7 kg to grams. Answer: _____

C2. Convert 8 L to millilitres. Answer: _____

C3. Convert 9 kg to grams. Answer: _____

C4. Convert 10 L to millilitres. Answer: _____

C5. Convert 11 kg to grams. Answer: _____

C6. Convert 12 L to millilitres. Answer: _____

C7. Convert 13 kg to grams. Answer: _____

C8. Convert 14 L to millilitres. Answer: _____

C9. Convert 15 kg to grams. Answer: _____

C10. Convert 16 L to millilitres. Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Convert 3 L to millilitres.

2. Work with a partner to explain why 4000 answers this: Convert 4 kg to grams.

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Convert 6 L to millilitres.

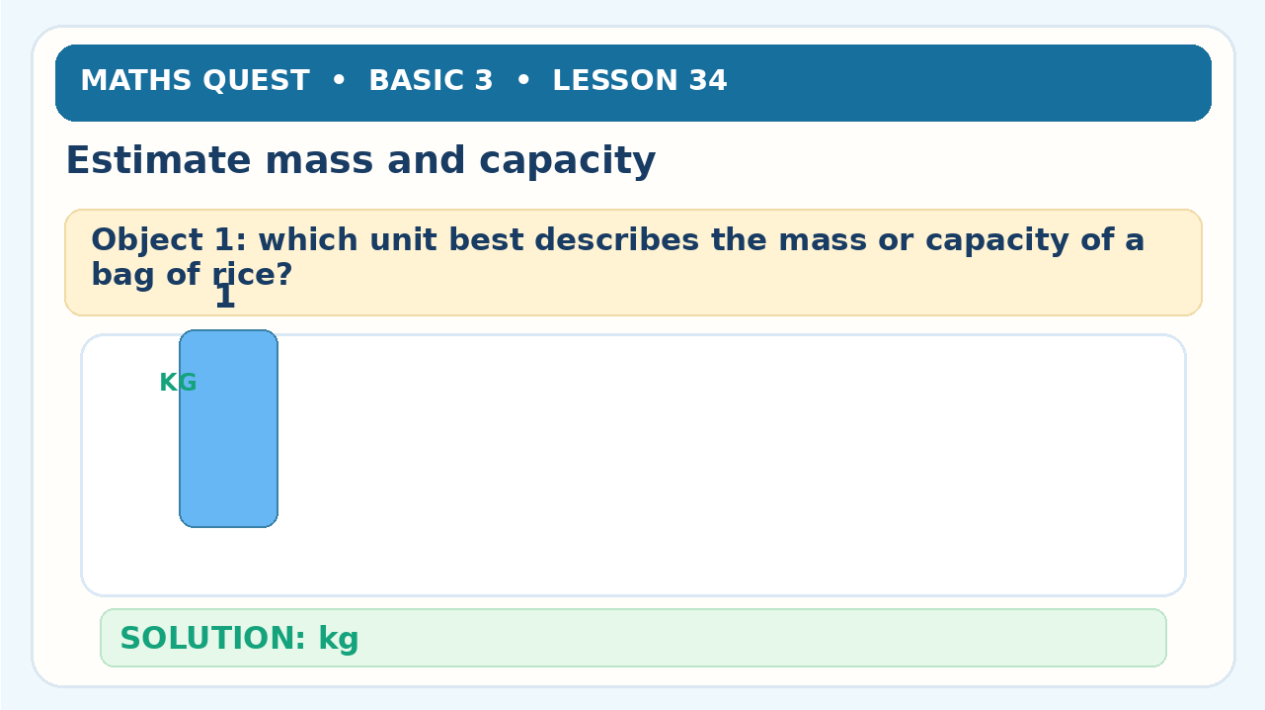
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 34 | Estimate mass and capacity

YOUR MISSION

Estimate masses and liquid capacity using familiar reference quantities.



LEARN IT - FOUR IMPORTANT IDEAS

1. A schoolbag may be measured in kilograms, while a small eraser may be measured in grams.

2. A bottle can hold litres or millilitres. Predict first, then check using suitable measuring equipment with an adult.

3. Worked example: Object 1: which unit best describes the mass or capacity of a bag of rice? Answer: kg. Choose mass units for weight and capacity units for liquid containers.

4. Check your understanding with another example: Object 2: which unit best describes the mass or capacity of a paperclip? The answer is g. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Object 1: which unit best describes the mass or capacity of a bag of rice?	kg — Choose mass units for weight and capacity units for liquid containers.
Object 2: which unit best describes the mass or capacity of a paperclip?	g — Choose mass units for weight and capacity units for liquid containers.

Try this	How it works
Object 3: which unit best describes the mass or capacity of a bucket of water?	L — Choose mass units for weight and capacity units for liquid containers.

TRY IT YOURSELF

P1. Object 1: which unit best describes the mass or capacity of a bag of rice? Answer: _____

P2. Object 2: which unit best describes the mass or capacity of a paperclip? Answer: _____

P3. Object 3: which unit best describes the mass or capacity of a bucket of water? Answer: _____

P4. Object 4: which unit best describes the mass or capacity of a tea spoon of water? Answer: _____

P5. Object 5: which unit best describes the mass or capacity of a schoolbag? Answer: _____

CHALLENGE PRACTICE

C1. Object 6: which unit best describes the mass or capacity of a bottle cap? Answer: _____

C2. Object 7: which unit best describes the mass or capacity of a bag of rice? Answer: _____

C3. Object 8: which unit best describes the mass or capacity of a paperclip? Answer: _____

C4. Object 9: which unit best describes the mass or capacity of a bucket of water? Answer: _____

C5. Object 10: which unit best describes the mass or capacity of a tea spoon of water? Answer: _____

C6. Object 11: which unit best describes the mass or capacity of a schoolbag? Answer: _____

C7. Object 12: which unit best describes the mass or capacity of a bottle cap? Answer: _____

C8. Object 13: which unit best describes the mass or capacity of a bag of rice? Answer: _____

C9. Object 14: which unit best describes the mass or capacity of a paperclip? Answer: _____

C10. Object 15: which unit best describes the mass or capacity of a bucket of water? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Object 3: which unit best describes the mass or capacity of a bucket of water?
2. Work with a partner to explain why mL answers this: Object 4: which unit best describes the mass or capacity of a tea spoon of water?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Object 5: which unit best describes the mass or capacity of a schoolbag?

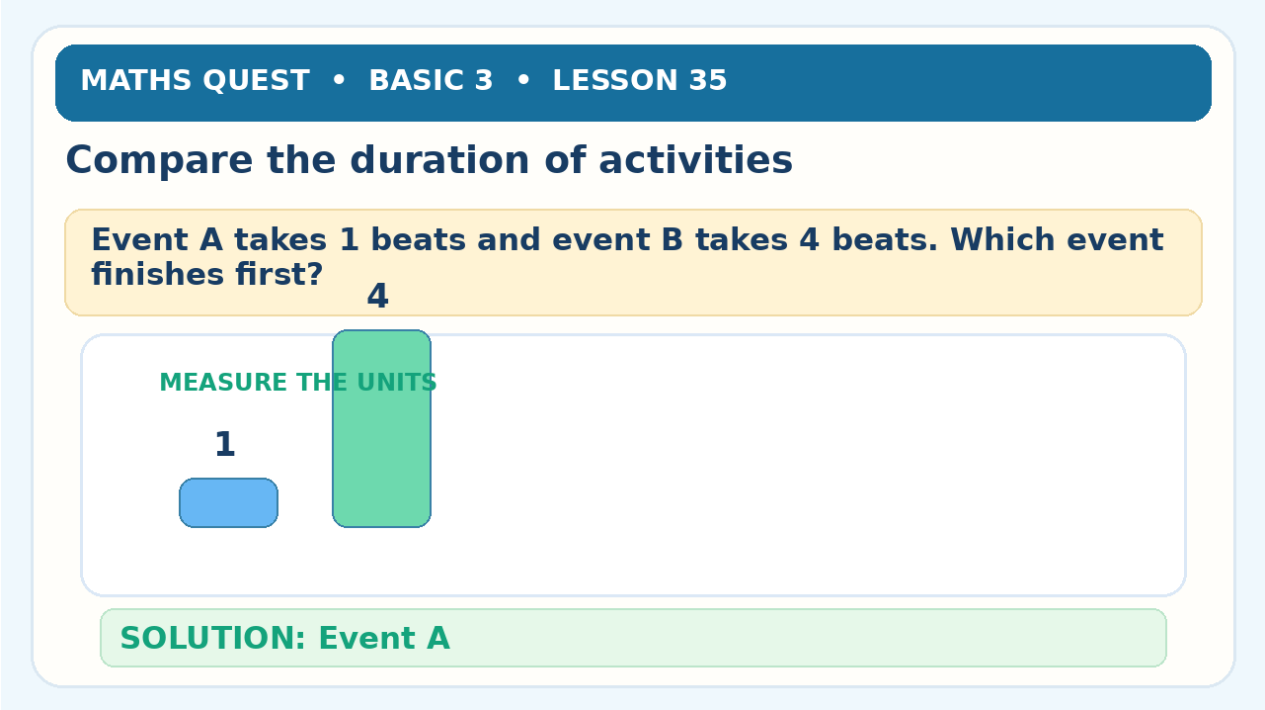
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 35 | Compare the duration of activities

YOUR MISSION

Use repeated units such as claps or swings to compare how long activities take.



LEARN IT - FOUR IMPORTANT IDEAS

1. Duration is how long an activity lasts. Choose the same repeated unit when comparing two events.

2. A five-clap activity lasts less than an eight-clap activity when the claps are evenly spaced.

3. Worked example: Event A takes 1 beats and event B takes 4 beats. Which event finishes first? Answer: Event A. The event with 1 beats is shorter than the event with 4 beats.

4. Check your understanding with another example: Event A takes 2 beats and event B takes 6 beats. Which event finishes last? The answer is Event B. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Event A takes 1 beats and event B takes 4 beats. Which event finishes first?	Event A — The event with 1 beats is shorter than the event with 4 beats.
Event A takes 2 beats and event B takes 6 beats. Which event finishes last?	Event B — The event with 2 beats is shorter than the event with 6 beats.

Try this	How it works
Event A takes 3 beats and event B takes 8 beats. Which event finishes first?	Event A — The event with 3 beats is shorter than the event with 8 beats.

TRY IT YOURSELF

P1. Event A takes 1 beats and event B takes 4 beats. Which event finishes first? Answer: _____

P2. Event A takes 2 beats and event B takes 6 beats. Which event finishes last? Answer: _____

P3. Event A takes 3 beats and event B takes 8 beats. Which event finishes first? Answer: _____

P4. Event A takes 4 beats and event B takes 10 beats. Which event finishes last? Answer: _____

P5. Event A takes 5 beats and event B takes 8 beats. Which event finishes first? Answer: _____

CHALLENGE PRACTICE

C1. Event A takes 6 beats and event B takes 10 beats. Which event finishes last? Answer: _____

C2. Event A takes 7 beats and event B takes 12 beats. Which event finishes first? Answer: _____

C3. Event A takes 8 beats and event B takes 14 beats. Which event finishes last? Answer: _____

C4. Event A takes 9 beats and event B takes 12 beats. Which event finishes first? Answer: _____

C5. Event A takes 10 beats and event B takes 14 beats. Which event finishes last? Answer: _____

C6. Event A takes 11 beats and event B takes 16 beats. Which event finishes first? Answer: _____

C7. Event A takes 12 beats and event B takes 18 beats. Which event finishes last? Answer: _____

C8. Event A takes 13 beats and event B takes 16 beats. Which event finishes first? Answer: _____

C9. Event A takes 14 beats and event B takes 18 beats. Which event finishes last? Answer: _____

C10. Event A takes 15 beats and event B takes 20 beats. Which event finishes first? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Event A takes 3 beats and event B takes 8 beats. Which event finishes first?
2. Work with a partner to explain why Event B answers this: Event A takes 4 beats and event B takes 10 beats. Which event finishes last?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Event A takes 5 beats and event B takes 8 beats. Which event finishes first?

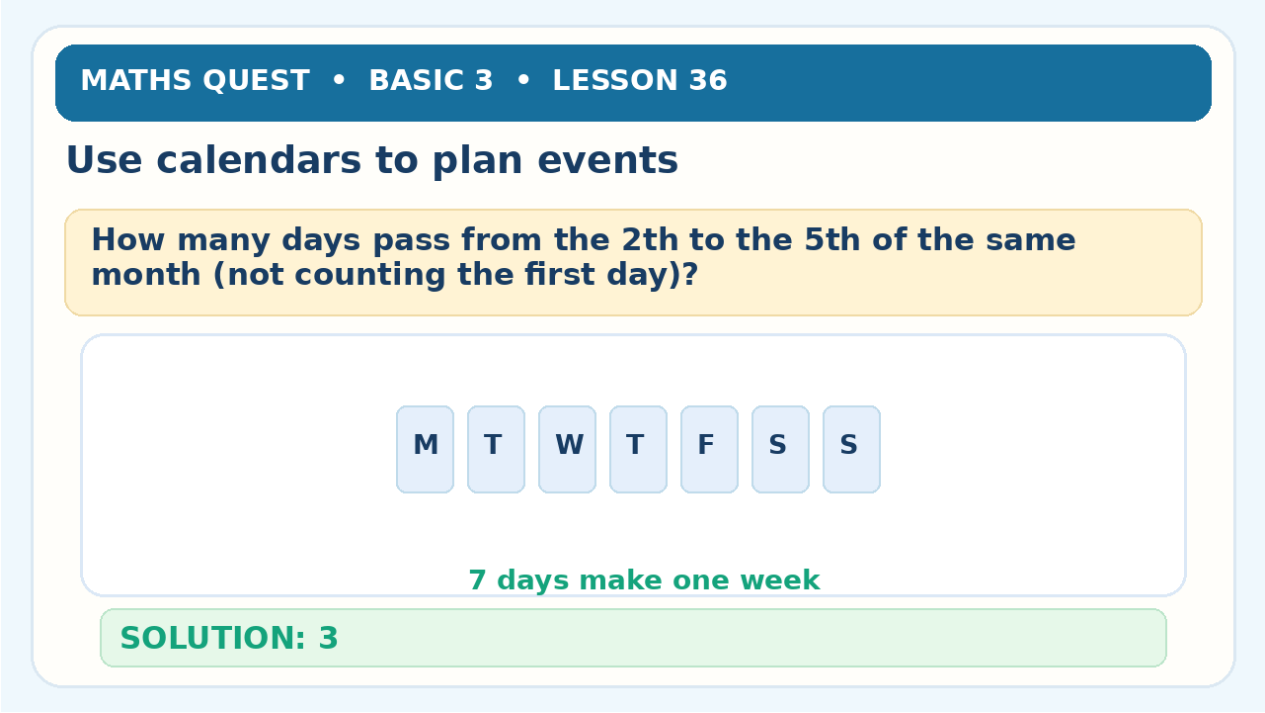
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 36 | Use calendars to plan events

YOUR MISSION

Read and order dates; count days, weeks and months between events.



LEARN IT - FOUR IMPORTANT IDEAS

1. A calendar shows dates in weeks and months. Seven days make a week.

2. To count days between dates, move one day at a time and decide whether your count includes the starting day.

3. Worked example: How many days pass from the 2th to the 5th of the same month (not counting the first day)? Answer: 3. Count forward 3 days.

4. Check your understanding with another example: How many days pass from the 3th to the 7th of the same month (not counting the first day)? The answer is 4. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
How many days pass from the 2th to the 5th of the same month (not counting the first day)?	3 — Count forward 3 days.

Try this	How it works
How many days pass from the 3th to the 7th of the same month (not counting the first day)?	4 — Count forward 4 days.
How many days pass from the 4th to the 9th of the same month (not counting the first day)?	5 — Count forward 5 days.

TRY IT YOURSELF

P1. How many days pass from the 2th to the 5th of the same month (not counting the first day)?
Answer: _____

P2. How many days pass from the 3th to the 7th of the same month (not counting the first day)?
Answer: _____

P3. How many days pass from the 4th to the 9th of the same month (not counting the first day)?
Answer: _____

P4. How many days pass from the 5th to the 11th of the same month (not counting the first day)?
Answer: _____

P5. How many days pass from the 6th to the 13th of the same month (not counting the first day)?
Answer: _____

CHALLENGE PRACTICE

C1. How many days pass from the 7th to the 15th of the same month (not counting the first day)?
Answer: _____

C2. How many days pass from the 8th to the 17th of the same month (not counting the first day)?
Answer: _____

C3. How many days pass from the 9th to the 19th of the same month (not counting the first day)?
Answer: _____

C4. How many days pass from the 10th to the 13th of the same month (not counting the first day)?
Answer: _____

C5. How many days pass from the 11th to the 15th of the same month (not counting the first day)?
Answer: _____

C6. How many days pass from the 12th to the 17th of the same month (not counting the first day)?
Answer: _____

C7. How many days pass from the 13th to the 19th of the same month (not counting the first day)?
Answer: _____

C8. How many days pass from the 14th to the 21th of the same month (not counting the first day)?
Answer: _____

C9. How many days pass from the 15th to the 23th of the same month (not counting the first day)?
Answer: _____

C10. How many days pass from the 16th to the 25th of the same month (not counting the first day)?
Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: How many days pass from the 4th to the 9th of the same month (not counting the first day)?

2. Work with a partner to explain why 6 answers this: How many days pass from the 5th to the 11th of the same month (not counting the first day)?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: How many days pass from the 6th to the 13th of the same month (not counting the first day)?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 37 | Convert seconds, minutes and hours

YOUR MISSION

Relate 60 seconds to a minute, 60 minutes to an hour and 24 hours to a day.

MATHS QUEST • BASIC 3 • LESSON 37

Convert seconds, minutes and hours

Time card 1: how many minutes are in 1 hours?



SOLUTION: 60

LEARN IT - FOUR IMPORTANT IDEAS

1. A minute contains sixty seconds. An hour contains sixty minutes.

2. There are twenty-four hours in a day. Multiply or divide by sixty when converting complete minutes and hours.

3. Worked example: Time card 1: how many minutes are in 1 hours? Answer: 60. Multiply by 60.

4. Check your understanding with another example: Time card 2: how many seconds are in 2 minutes? The answer is 120. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
Time card 1: how many minutes are in 1 hours?	60 — Multiply by 60.
Time card 2: how many seconds are in 2 minutes?	120 — Multiply by 60.

Try this	How it works
Time card 3: how many minutes are in 3 hours?	180 — Multiply by 60.

TRY IT YOURSELF

P1. Time card 1: how many minutes are in 1 hours? Answer: _____

P2. Time card 2: how many seconds are in 2 minutes? Answer: _____

P3. Time card 3: how many minutes are in 3 hours? Answer: _____

P4. Time card 4: how many seconds are in 4 minutes? Answer: _____

P5. Time card 5: how many minutes are in 5 hours? Answer: _____

CHALLENGE PRACTICE

C1. Time card 6: how many seconds are in 6 minutes? Answer: _____

C2. Time card 7: how many minutes are in 7 hours? Answer: _____

C3. Time card 8: how many seconds are in 8 minutes? Answer: _____

C4. Time card 9: how many minutes are in 1 hours? Answer: _____

C5. Time card 10: how many seconds are in 2 minutes? Answer: _____

C6. Time card 11: how many minutes are in 3 hours? Answer: _____

C7. Time card 12: how many seconds are in 4 minutes? Answer: _____

C8. Time card 13: how many minutes are in 5 hours? Answer: _____

C9. Time card 14: how many seconds are in 6 minutes? Answer: _____

C10. Time card 15: how many minutes are in 7 hours? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: Time card 3: how many minutes are in 3 hours?
2. Work with a partner to explain why 240 answers this: Time card 4: how many seconds are in 4 minutes?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: Time card 5: how many minutes are in 5 hours?

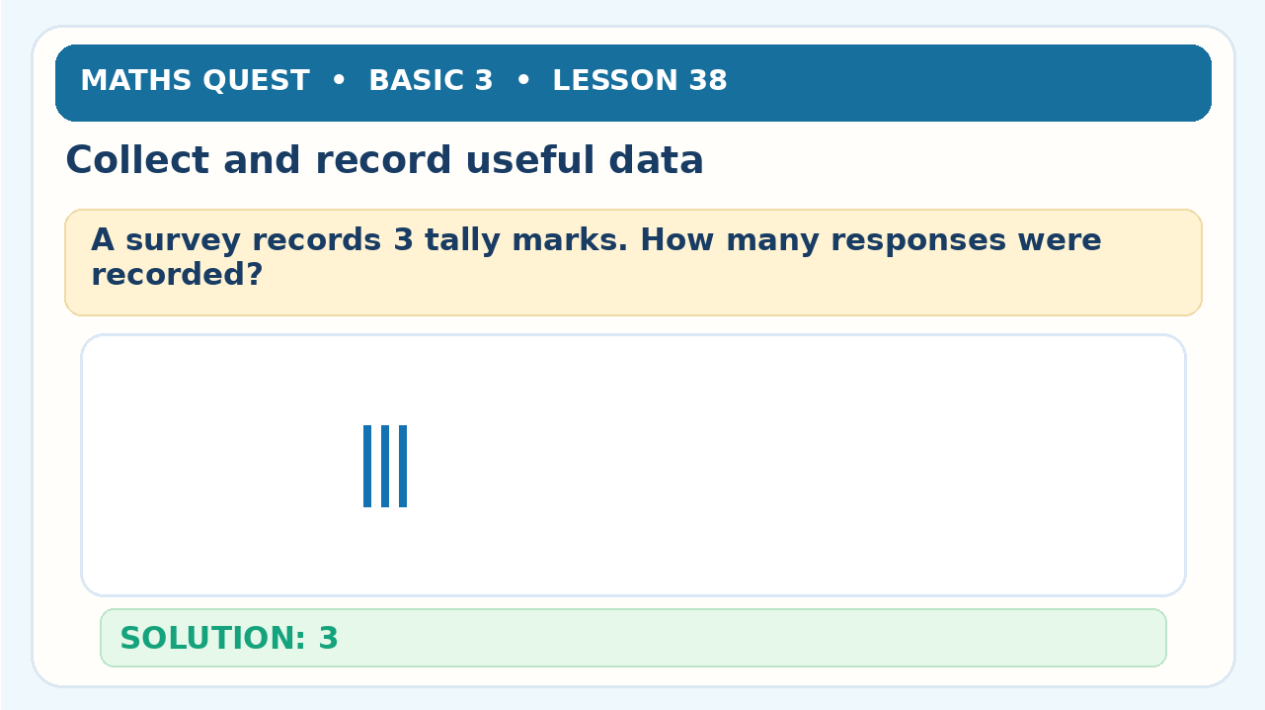
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 38 | Collect and record useful data

YOUR MISSION

Ask questions and record responses using tables and tally marks.



LEARN IT - FOUR IMPORTANT IDEAS

1. Data are facts gathered to answer a question. Put responses into clear categories before counting.

2. A tally mark represents one response. A diagonal fifth mark makes a group of five that is easier to total.

3. Worked example: A survey records 3 tally marks. How many responses were recorded? Answer: 3. Each tally counts one response.

4. Check your understanding with another example: A survey records 4 tally marks. How many responses were recorded? The answer is 4. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
A survey records 3 tally marks. How many responses were recorded?	3 — Each tally counts one response.
A survey records 4 tally marks. How many responses were recorded?	4 — Each tally counts one response.

Try this	How it works
A survey records 5 tally marks. How many responses were recorded?	5 — Each tally counts one response.

TRY IT YOURSELF

P1. A survey records 3 tally marks. How many responses were recorded? Answer: _____

P2. A survey records 4 tally marks. How many responses were recorded? Answer: _____

P3. A survey records 5 tally marks. How many responses were recorded? Answer: _____

P4. A survey records 6 tally marks. How many responses were recorded? Answer: _____

P5. A survey records 7 tally marks. How many responses were recorded? Answer: _____

CHALLENGE PRACTICE

C1. A survey records 8 tally marks. How many responses were recorded? Answer: _____

C2. A survey records 9 tally marks. How many responses were recorded? Answer: _____

C3. A survey records 10 tally marks. How many responses were recorded? Answer: _____

C4. A survey records 11 tally marks. How many responses were recorded? Answer: _____

C5. A survey records 12 tally marks. How many responses were recorded? Answer: _____

C6. A survey records 13 tally marks. How many responses were recorded? Answer: _____

C7. A survey records 14 tally marks. How many responses were recorded? Answer: _____

C8. A survey records 15 tally marks. How many responses were recorded? Answer: _____

C9. A survey records 16 tally marks. How many responses were recorded? Answer: _____

C10. A survey records 17 tally marks. How many responses were recorded? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: A survey records 5 tally marks. How many responses were recorded?
2. Work with a partner to explain why 6 answers this: A survey records 6 tally marks. How many responses were recorded?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: A survey records 7 tally marks. How many responses were recorded?

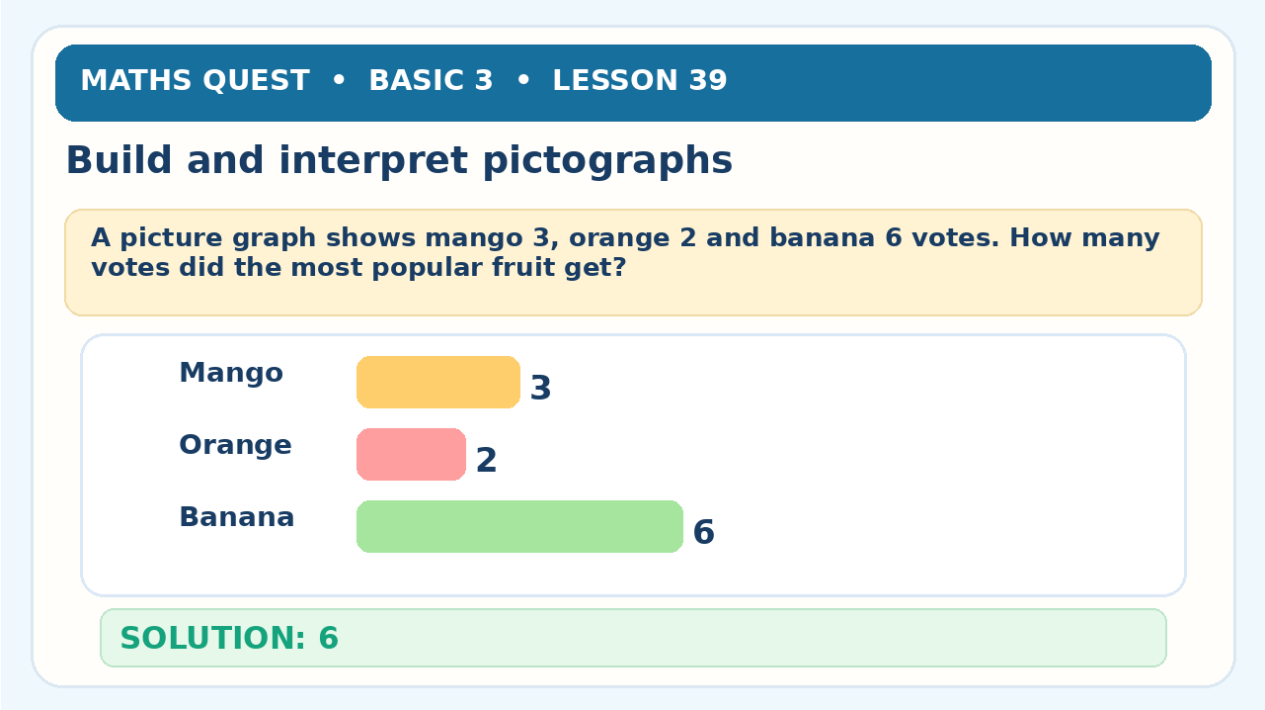
THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Quest 39 | Build and interpret pictographs

YOUR MISSION

Create pictographs and read data to answer comparison questions.



LEARN IT - FOUR IMPORTANT IDEAS

1. A pictograph uses pictures to represent data. Read its key before counting symbols.

2. The key may say one picture equals two children. Multiply the number of pictures by the key value to get the data total.

3. Worked example: A picture graph shows mango 3, orange 2 and banana 6 votes. How many votes did the most popular fruit get? Answer: 6. Compare the three totals and choose the largest.

4. Check your understanding with another example: A picture graph shows mango 4, orange 3 and banana 7 votes. How many votes did the most popular fruit get? The answer is 7. Describe each step in your own words.

WORKED EXAMPLES

Try this	How it works
A picture graph shows mango 3, orange 2 and banana 6 votes. How many votes did the most popular fruit get?	6 — Compare the three totals and choose the largest.

Try this	How it works
A picture graph shows mango 4, orange 3 and banana 7 votes. How many votes did the most popular fruit get?	7 — Compare the three totals and choose the largest.
A picture graph shows mango 5, orange 4 and banana 4 votes. How many votes did the most popular fruit get?	5 — Compare the three totals and choose the largest.

TRY IT YOURSELF

P1. A picture graph shows mango 3, orange 2 and banana 6 votes. How many votes did the most popular fruit get? Answer: _____

P2. A picture graph shows mango 4, orange 3 and banana 7 votes. How many votes did the most popular fruit get? Answer: _____

P3. A picture graph shows mango 5, orange 4 and banana 4 votes. How many votes did the most popular fruit get? Answer: _____

P4. A picture graph shows mango 6, orange 2 and banana 5 votes. How many votes did the most popular fruit get? Answer: _____

P5. A picture graph shows mango 7, orange 3 and banana 6 votes. How many votes did the most popular fruit get? Answer: _____

CHALLENGE PRACTICE

C1. A picture graph shows mango 3, orange 4 and banana 7 votes. How many votes did the most popular fruit get? Answer: _____

C2. A picture graph shows mango 4, orange 2 and banana 4 votes. How many votes did the most popular fruit get? Answer: _____

C3. A picture graph shows mango 5, orange 3 and banana 5 votes. How many votes did the most popular fruit get? Answer: _____

C4. A picture graph shows mango 6, orange 4 and banana 6 votes. How many votes did the most popular fruit get? Answer: _____

C5. A picture graph shows mango 7, orange 2 and banana 7 votes. How many votes did the most popular fruit get? Answer: _____

C6. A picture graph shows mango 3, orange 3 and banana 4 votes. How many votes did the most popular fruit get? Answer: _____

C7. A picture graph shows mango 4, orange 4 and banana 5 votes. How many votes did the most popular fruit get? Answer: _____

C8. A picture graph shows mango 5, orange 2 and banana 6 votes. How many votes did the most popular fruit get? Answer: _____

C9. A picture graph shows mango 6, orange 3 and banana 7 votes. How many votes did the most popular fruit get? Answer: _____

C10. A picture graph shows mango 7, orange 4 and banana 4 votes. How many votes did the most popular fruit get? Answer: _____

REAL-LIFE INVESTIGATION

1. Model the idea using paper, pencils or bottle tops: A picture graph shows mango 5, orange 4 and banana 4 votes. How many votes did the most popular fruit get?

2. Work with a partner to explain why 6 answers this: A picture graph shows mango 6, orange 2 and banana 5 votes. How many votes did the most popular fruit get?

AT HOME

Show a parent or trusted adult how to solve this question using objects or a drawing: A picture graph shows mango 7, orange 3 and banana 6 votes. How many votes did the most popular fruit get?

THINK BACK

Tell someone: what did you learn? Which example can you explain without looking at the answer?

Answer key for learner practice

Quest 01: Count and estimate to 10,000

Play 1: 70 • Play 2: 250 • Play 3: 500 • Play 4: 1500 • Play 5: 3000

Challenge 1: 1070 • Challenge 2: 1250 • Challenge 3: 1500 • Challenge 4: 2500 • Challenge 5: 4000 • Challenge 6: 2070 • Challenge 7: 2250 • Challenge 8: 2500 • Challenge 9: 3500 • Challenge 10: 5000

Quest 02: Find neighbours on large number charts

Play 1: 222 • Play 2: 251 • Play 3: 293 • Play 4: 304 • Play 5: 346

Challenge 1: 375 • Challenge 2: 417 • Challenge 3: 428 • Challenge 4: 470 • Challenge 5: 499 • Challenge 6: 541 • Challenge 7: 552 • Challenge 8: 594 • Challenge 9: 623 • Challenge 10: 665

Quest 03: Build four-digit numbers

Play 1: 4000 • Play 2: 800 • Play 3: 30 • Play 4: 7 • Play 5: 7000

Challenge 1: 200 • Challenge 2: 0 • Challenge 3: 9 • Challenge 4: 1000 • Challenge 5: 900 • Challenge 6: 0 • Challenge 7: 3 • Challenge 8: 6000 • Challenge 9: 800 • Challenge 10: 80

Quest 04: Compare and order numbers to 10,000

Play 1: < • Play 2: > • Play 3: = • Play 4: < • Play 5: >

Challenge 1: < • Challenge 2: > • Challenge 3: = • Challenge 4: < • Challenge 5: > • Challenge 6: < • Challenge 7: > • Challenge 8: = • Challenge 9: < • Challenge 10: >

Quest 05: Move in opposite directions

Play 1: right • Play 2: down • Play 3: backward • Play 4: south • Play 5: west

Challenge 1: left • Challenge 2: up • Challenge 3: forward • Challenge 4: north • Challenge 5: east • Challenge 6: right • Challenge 7: down • Challenge 8: backward • Challenge 9: south • Challenge 10: west

Quest 06: Understand positive and negative situations

Play 1: 1 • Play 2: -2 • Play 3: 3 • Play 4: -4 • Play 5: 5

Challenge 1: -6 • Challenge 2: 7 • Challenge 3: -8 • Challenge 4: 9 • Challenge 5: -10 • Challenge 6: 1 • Challenge 7: -2 • Challenge 8: 3 • Challenge 9: -4 • Challenge 10: 5

Quest 07: Read negative numbers to minus ten

Play 1: -9 • Play 2: -8 • Play 3: -7 • Play 4: -6 • Play 5: -5

Challenge 1: -4 • Challenge 2: -3 • Challenge 3: -2 • Challenge 4: -1 • Challenge 5: 0 • Challenge 6: -9 • Challenge 7: -8 • Challenge 8: -7 • Challenge 9: -6 • Challenge 10: -5

Quest 08: Use standard addition and subtraction

Play 1: 165 • Play 2: 107 • Play 3: 249 • Play 4: 139 • Play 5: 333

Challenge 1: 171 • Challenge 2: 417 • Challenge 3: 294 • Challenge 4: 410 • Challenge 5: 326 • Challenge 6: 494 • Challenge 7: 358 • Challenge 8: 578 • Challenge 9: 390 • Challenge 10: 571

Quest 09: Balance equal number statements

Play 1: Not equal • Play 2: Equal • Play 3: Equal • Play 4: Not equal • Play 5: Equal

Challenge 1: Equal • Challenge 2: Not equal • Challenge 3: Equal • Challenge 4: Equal • Challenge 5: Not equal • Challenge 6: Equal • Challenge 7: Equal • Challenge 8: Not equal • Challenge 9: Equal • Challenge 10: Equal

Quest 10: Mental strategies within 1,000

Play 1: 165 • Play 2: 207 • Play 3: 249 • Play 4: 291 • Play 5: 333

Challenge 1: 375 • Challenge 2: 417 • Challenge 3: 368 • Challenge 4: 410 • Challenge 5: 452 • Challenge 6: 494 • Challenge 7: 536 • Challenge 8: 578 • Challenge 9: 620 • Challenge 10: 571

Quest 11: Flexible addition strategies

Play 1: 165 • Play 2: 207 • Play 3: 249 • Play 4: 291 • Play 5: 333

Challenge 1: 375 • Challenge 2: 417 • Challenge 3: 368 • Challenge 4: 410 • Challenge 5: 452 • Challenge 6: 494 • Challenge 7: 536 • Challenge 8: 578 • Challenge 9: 620 • Challenge 10: 571

Quest 12: Flexible subtraction strategies

Play 1: 91 • Play 2: 107 • Play 3: 123 • Play 4: 139 • Play 5: 155

Challenge 1: 171 • Challenge 2: 187 • Challenge 3: 294 • Challenge 4: 310 • Challenge 5: 326 • Challenge 6: 342 • Challenge 7: 358 • Challenge 8: 374 • Challenge 9: 390 • Challenge 10: 497

Quest 13: Estimate sums and differences

Play 1: 170 • Play 2: 220 • Play 3: 260 • Play 4: 300 • Play 5: 340

Challenge 1: 390 • Challenge 2: 430 • Challenge 3: 380 • Challenge 4: 430 • Challenge 5: 460 • Challenge 6: 500 • Challenge 7: 540 • Challenge 8: 590 • Challenge 9: 630 • Challenge 10: 580

Quest 14: Explore commutative addition

Play 1: 165 • Play 2: 207 • Play 3: 249 • Play 4: 291 • Play 5: 333

Challenge 1: 375 • Challenge 2: 417 • Challenge 3: 368 • Challenge 4: 410 • Challenge 5: 452 • Challenge 6: 494 • Challenge 7: 536 • Challenge 8: 578 • Challenge 9: 620 • Challenge 10: 571

Quest 15: Multiplication as equal groups

Play 1: 1 • Play 2: 4 • Play 3: 9 • Play 4: 16 • Play 5: 25

Challenge 1: 2 • Challenge 2: 6 • Challenge 3: 12 • Challenge 4: 20 • Challenge 5: 5 • Challenge 6: 3 • Challenge 7: 8 • Challenge 8: 15 • Challenge 9: 4 • Challenge 10: 10

Quest 16: Multiplication arrays

Play 1: 1 • Play 2: 4 • Play 3: 9 • Play 4: 16 • Play 5: 25

Challenge 1: 2 • Challenge 2: 6 • Challenge 3: 12 • Challenge 4: 20 • Challenge 5: 5 • Challenge 6: 3 • Challenge 7: 8 • Challenge 8: 15 • Challenge 9: 4 • Challenge 10: 10

Quest 17: Division by sharing equally

Play 1: 2 • Play 2: 2 • Play 3: 2 • Play 4: 2 • Play 5: 3

Challenge 1: 3 • Challenge 2: 3 • Challenge 3: 3 • Challenge 4: 4 • Challenge 5: 4 • Challenge 6: 4 • Challenge 7: 4 • Challenge 8: 5 • Challenge 9: 5 • Challenge 10: 5

Quest 18: Division by repeated subtraction

Play 1: 2 • Play 2: 2 • Play 3: 2 • Play 4: 2 • Play 5: 3

Challenge 1: 3 • Challenge 2: 3 • Challenge 3: 3 • Challenge 4: 4 • Challenge 5: 4 • Challenge 6: 4 • Challenge 7: 4 • Challenge 8: 5 • Challenge 9: 5 • Challenge 10: 5

Quest 19: Division and multiplication families

Play 1: 2 • Play 2: 2 • Play 3: 2 • Play 4: 2 • Play 5: 3

Challenge 1: 3 • Challenge 2: 3 • Challenge 3: 3 • Challenge 4: 4 • Challenge 5: 4 • Challenge 6: 4 • Challenge 7: 4 • Challenge 8: 5 • Challenge 9: 5 • Challenge 10: 5

Quest 20: Understand unit fractions

Play 1: 1/2 • Play 2: 1/3 • Play 3: 1/4 • Play 4: 1/5 • Play 5: 1/6

Challenge 1: 1/2 • Challenge 2: 1/3 • Challenge 3: 1/4 • Challenge 4: 1/5 • Challenge 5: 1/6 • Challenge 6: 1/2 • Challenge 7: 1/3 • Challenge 8: 1/4 • Challenge 9: 1/5 • Challenge 10: 1/6

Quest 21: Fractions of collections

Play 1: 2 • Play 2: 2 • Play 3: 2 • Play 4: 2 • Play 5: 3

Challenge 1: 3 • Challenge 2: 3 • Challenge 3: 3 • Challenge 4: 4 • Challenge 5: 4 • Challenge 6: 4 • Challenge 7: 4 • Challenge 8: 5 • Challenge 9: 5 • Challenge 10: 5

Quest 22: Compare like-denominator fractions

Play 1: < • Play 2: < • Play 3: < • Play 4: < • Play 5: <

Challenge 1: < • Challenge 2: < • Challenge 3: < • Challenge 4: > • Challenge 5: < • Challenge 6: < • Challenge 7: > • Challenge 8: < • Challenge 9: < • Challenge 10: >

Quest 23: Use Ghanaian money denominations

Play 1: 3 • Play 2: 6 • Play 3: 15 • Play 4: 30 • Play 5: 41

Challenge 1: 102 • Challenge 2: 7 • Challenge 3: 14 • Challenge 4: 11 • Challenge 5: 22 • Challenge 6: 45 • Challenge 7: 110 • Challenge 8: 3 • Challenge 9: 6 • Challenge 10: 15

Quest 24: Investigate growing and shrinking patterns

Play 1: 46 • Play 2: 52 • Play 3: 58 • Play 4: 64 • Play 5: 46

Challenge 1: 46 • Challenge 2: 64 • Challenge 3: 70 • Challenge 4: 76 • Challenge 5: 82 • Challenge 6: 64 • Challenge 7: 64 • Challenge 8: 82 • Challenge 9: 88 • Challenge 10: 94

Quest 25: Discover faces, edges and vertices

Play 1: 6 • Play 2: 6 • Play 3: 5 • Play 4: 2 • Play 5: 0

Challenge 1: 1 • Challenge 2: 6 • Challenge 3: 6 • Challenge 4: 5 • Challenge 5: 2 • Challenge 6: 0 • Challenge 7: 1 • Challenge 8: 6 • Challenge 9: 6 • Challenge 10: 5

Quest 26: Draw and recognise angles

Play 1: acute • Play 2: acute • Play 3: acute • Play 4: acute • Play 5: acute

Challenge 1: acute • Challenge 2: acute • Challenge 3: acute • Challenge 4: acute • Challenge 5: acute • Challenge 6: right • Challenge 7: obtuse • Challenge 8: obtuse • Challenge 9: obtuse • Challenge 10: obtuse

Quest 27: Compare angles with a square corner

Play 1: smaller • Play 2: smaller • Play 3: smaller • Play 4: smaller • Play 5: smaller

Challenge 1: smaller • Challenge 2: smaller • Challenge 3: equal • Challenge 4: larger • Challenge 5: larger • Challenge 6: larger • Challenge 7: larger • Challenge 8: larger • Challenge 9: larger • Challenge 10: larger

Quest 28: Identify quadrilaterals

Play 1: rectangle • Play 2: square • Play 3: rhombus • Play 4: parallelogram • Play 5: trapezoid

Challenge 1: rectangle • Challenge 2: square • Challenge 3: rhombus • Challenge 4: parallelogram • Challenge 5: trapezoid • Challenge 6: rectangle • Challenge 7: square • Challenge 8: rhombus • Challenge 9: parallelogram • Challenge 10: trapezoid

Quest 29: Measure distances on number lines

Play 1: 5 • Play 2: 6 • Play 3: 7 • Play 4: 8 • Play 5: 9

Challenge 1: 10 • Challenge 2: 11 • Challenge 3: 5 • Challenge 4: 6 • Challenge 5: 7 • Challenge 6: 8 • Challenge 7: 9 • Challenge 8: 10 • Challenge 9: 11 • Challenge 10: 5

Quest 30: Metres and centimetres

Play 1: 100 • Play 2: 200 • Play 3: 300 • Play 4: 400 • Play 5: 500

Challenge 1: 600 • Challenge 2: 700 • Challenge 3: 800 • Challenge 4: 900 • Challenge 5: 1000 • Challenge 6: 1100 • Challenge 7: 1200 • Challenge 8: 1300 • Challenge 9: 1400 • Challenge 10: 1500

Quest 31: Choose suitable length units

Play 1: cm • Play 2: m • Play 3: cm • Play 4: m • Play 5: cm

Challenge 1: m • Challenge 2: cm • Challenge 3: m • Challenge 4: cm • Challenge 5: m • Challenge 6: cm • Challenge 7: m • Challenge 8: cm • Challenge 9: m • Challenge 10: cm

Quest 32: Estimate and measure perimeter

Play 1: 10 • Play 2: 16 • Play 3: 22 • Play 4: 28 • Play 5: 20

Challenge 1: 26 • Challenge 2: 32 • Challenge 3: 24 • Challenge 4: 14 • Challenge 5: 20 • Challenge 6: 26 • Challenge 7: 18 • Challenge 8: 24 • Challenge 9: 30 • Challenge 10: 22

Quest 33: Grams, kilograms, millilitres and litres

Play 1: 2000 • Play 2: 5000 • Play 3: 3000 • Play 4: 4000 • Play 5: 6000

Challenge 1: 7000 • Challenge 2: 8000 • Challenge 3: 9000 • Challenge 4: 10000 • Challenge 5: 11000 • Challenge 6: 12000 • Challenge 7: 13000 • Challenge 8: 14000 • Challenge 9: 15000 • Challenge 10: 16000

Quest 34: Estimate mass and capacity

Play 1: kg • Play 2: g • Play 3: L • Play 4: mL • Play 5: kg

Challenge 1: g • Challenge 2: kg • Challenge 3: g • Challenge 4: L • Challenge 5: mL • Challenge 6: kg • Challenge 7: g • Challenge 8: kg • Challenge 9: g • Challenge 10: L

Quest 35: Compare the duration of activities

Play 1: Event A • Play 2: Event B • Play 3: Event A • Play 4: Event B • Play 5: Event A

Challenge 1: Event B • Challenge 2: Event A • Challenge 3: Event B • Challenge 4: Event A • Challenge 5: Event B • Challenge 6: Event A • Challenge 7: Event B • Challenge 8: Event A • Challenge 9: Event B • Challenge 10: Event A

Quest 36: Use calendars to plan events

Play 1: 3 • Play 2: 4 • Play 3: 5 • Play 4: 6 • Play 5: 7

Challenge 1: 8 • Challenge 2: 9 • Challenge 3: 10 • Challenge 4: 3 • Challenge 5: 4 • Challenge 6: 5 • Challenge 7: 6 • Challenge 8: 7 • Challenge 9: 8 • Challenge 10: 9

Quest 37: Convert seconds, minutes and hours

Play 1: 60 • Play 2: 120 • Play 3: 180 • Play 4: 240 • Play 5: 300

Challenge 1: 360 • Challenge 2: 420 • Challenge 3: 480 • Challenge 4: 60 • Challenge 5: 120 • Challenge 6: 180 • Challenge 7: 240 • Challenge 8: 300 • Challenge 9: 360 • Challenge 10: 420

Quest 38: Collect and record useful data

Play 1: 3 • Play 2: 4 • Play 3: 5 • Play 4: 6 • Play 5: 7

Challenge 1: 8 • Challenge 2: 9 • Challenge 3: 10 • Challenge 4: 11 • Challenge 5: 12 • Challenge 6: 13 • Challenge 7: 14 • Challenge 8: 15 • Challenge 9: 16 • Challenge 10: 17

Quest 39: Build and interpret pictographs

Play 1: 6 • Play 2: 7 • Play 3: 5 • Play 4: 6 • Play 5: 7

Challenge 1: 7 • Challenge 2: 4 • Challenge 3: 5 • Challenge 4: 6 • Challenge 5: 7 • Challenge 6: 4 • Challenge 7: 5 • Challenge 8: 6 • Challenge 9: 7 • Challenge 10: 7